

Visualizing Data using t-SNE

Laurens van der Maaten

LVDMAATEN@GMAIL.COM

TiCC

Tilburg University

P.O. Box 90153, 5000 LE Tilburg, The Netherlands

Geoffrey Hinton

HINTON@CS.TORONTO.EDU

Department of Computer Science

University of Toronto

6 King's College Road, M5S 3G4 Toronto, ON, Canada

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Abstract

We present a new technique called “t-SNE” that visualizes high-dimensional data by giving each datapoint a location in a two or three-dimensional map. The technique is a variation of Stochastic Neighbor Embedding (Hinton and Roweis, 2002) that is much easier to optimize, and produces significantly better visualizations by reducing the tendency to crowd points together in the center of the map. t-SNE is better than existing techniques at creating a single map that reveals structure at many different scales. This is particularly important for high-dimensional data that lie on several different, but related, low-dimensional manifolds, such as images of objects from multiple classes seen from multiple viewpoints. For visualizing the structure of very large data sets, we show how t-SNE can use random walks on neighborhood graphs to allow the implicit structure of all of the data to influence the way in which a subset of the data is displayed. We illustrate the performance of t-SNE on a wide variety of data sets and compare it with many other non-parametric visualization techniques, including Sammon mapping, Isomap, and Locally Linear Embedding. The visualizations produced by t-SNE are significantly better than those produced by the other techniques on almost all of the data sets.

Keywords: visualization, dimensionality reduction, manifold learning, embedding algorithms, multidimensional scaling

1. Introduction

Visualization of high-dimensional data is an important problem in many different domains, and deals with data of widely varying dimensionality. Cell nuclei that are relevant to breast cancer, for example, are described by approximately 30 variables (Street et al., 1993), whereas the pixel intensity vectors used to represent images or the word-count vectors used to represent documents typically have thousands of dimensions. Over the last few decades, a variety of techniques for the visualization of such high-dimensional data have been proposed, many of which are reviewed by de Oliveira and Levkowitz (2003). Important techniques include iconographic displays such as Chernoff faces (Chernoff, 1973), pixel-based techniques (Keim, 2000), and techniques that represent the dimensions in the data as vertices in a graph (Battista et al., 1994). Most of these techniques simply provide tools to display more than two data dimensions, and leave the interpretation of the

使用 t-SNE 进行数据可视化

Laurens van der Maaten @.
TiCC
蒂尔堡大学 邮政信箱 90153,5000 LE 蒂尔堡, 荷兰

杰弗里·辛顿 @..
多伦多大学计算机科学系 6 King's College Road, M5S
3G4 多伦多, 加拿大安大略省

编辑: 约书亚·本吉奥

抽象

我们提出了一种名为“t-SNE”的新技术, 通过为每个数据点在二维或三维地图上赋予一个位置来可视化高维数据。该技术是随机邻居嵌入 (Stochastic Neighbor Embedding, Hinton 和 Roweis, 2002) 的一个变体, 更容易优化, 并通过减少地图中心点聚集的倾向, 显著提升了可视化效果。t-SNE 在创建单一地图上, 能够揭示多种尺度结构, 优于现有技术。这对于存在多个不同但相关的低维流形的高维数据尤为重要, 例如从多个视角观察到的多类物体图像。为了可视化超大型数据集的结构, 我们展示了 t-SNE 如何利用邻域图上的随机漫步, 使所有数据的隐式结构影响部分数据的显示方式。我们展示了 t-SNE 在多种数据集上的表现, 并与许多其他非参数可视化技术进行比较, 包括萨蒙映射、等值映射和局部线性嵌入。t-SNE 在几乎所有数据集上产生的可视化效果明显优于其他技术。

关键词: 可视化、降维、流形学习、嵌入算法、多维缩放

1. 引言

高维数据的可视化是许多不同领域中的重要问题, 涉及维度差异很大的数据。例如, 与乳腺癌相关的细胞核大约由 30 个变量描述 (Street 等, 1993), 而用于表示图像的像素强度矢量或用于表示文档的字数矢量通常具有数千个维度。在过去几十年里, 提出了多种用于可视化此类高维数据的技术, 其中许多由 de Oliveira 和 Levkowitz (2003 年) 进行了综述。重要技术包括图像显示, 如切尔诺夫面 (Chernoff, 1973)、基于像素的技术 (Keim, 2000), 以及将数据维度表示为图中的顶点的技术 (Battista 等, 1994)。大多数这些技术仅提供显示多个数据维度的工具, 并保留对

data to the human observer. This severely limits the applicability of these techniques to real-world data sets that contain thousands of high-dimensional datapoints.

In contrast to the visualization techniques discussed above, dimensionality reduction methods convert the high-dimensional data set $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$ into two or three-dimensional data $\mathcal{Y} = \{y_1, y_2, \dots, y_n\}$ that can be displayed in a scatterplot. In the paper, we refer to the low-dimensional data representation $\mathcal{Y}$ as a map, and to the low-dimensional representations y_i of individual datapoints as map points. The aim of dimensionality reduction is to preserve as much of the significant structure of the high-dimensional data as possible in the low-dimensional map. Various techniques for this problem have been proposed that differ in the type of structure they preserve. Traditional dimensionality reduction techniques such as Principal Components Analysis (PCA; Hotelling, 1933) and classical multidimensional scaling (MDS; Torgerson, 1952) are linear techniques that focus on keeping the low-dimensional representations of dissimilar datapoints far apart. For high-dimensional data that lies on or near a low-dimensional, non-linear manifold it is usually more important to keep the low-dimensional representations of very similar datapoints close together, which is typically not possible with a linear mapping.

A large number of nonlinear dimensionality reduction techniques that aim to preserve the local structure of data have been proposed, many of which are reviewed by Lee and Verleysen (2007). In particular, we mention the following seven techniques: (1) Sammon mapping (Sammon, 1969), (2) curvilinear components analysis (CCA; Demartines and Hérault, 1997), (3) Stochastic Neighbor Embedding (SNE; Hinton and Roweis, 2002), (4) Isomap (Tenenbaum et al., 2000), (5) Maximum Variance Unfolding (MVU; Weinberger et al., 2004), (6) Locally Linear Embedding (LLE; Roweis and Saul, 2000), and (7) Laplacian Eigenmaps (Belkin and Niyogi, 2002). Despite the strong performance of these techniques on artificial data sets, they are often not very successful at visualizing real, high-dimensional data. In particular, most of the techniques are not capable of retaining both the local and the global structure of the data in a single map. For instance, a recent study reveals that even a semi-supervised variant of MVU is not capable of separating handwritten digits into their natural clusters (Song et al., 2007).

In this paper, we describe a way of converting a high-dimensional data set into a matrix of pairwise similarities and we introduce a new technique, called “t-SNE”, for visualizing the resulting similarity data. t-SNE is capable of capturing much of the local structure of the high-dimensional data very well, while also revealing global structure such as the presence of clusters at several scales. We illustrate the performance of t-SNE by comparing it to the seven dimensionality reduction techniques mentioned above on five data sets from a variety of domains. Because of space limitations, most of the $(7 + 1) \times 5 = 40$ maps are presented in the supplemental material, but the maps that we present in the paper are sufficient to demonstrate the superiority of t-SNE.

The outline of the paper is as follows. In Section 2, we outline SNE as presented by Hinton and Roweis (2002), which forms the basis for t-SNE. In Section 3, we present t-SNE, which has two important differences from SNE. In Section 4, we describe the experimental setup and the results of our experiments. Subsequently, Section 5 shows how t-SNE can be modified to visualize real-world data sets that contain many more than 10,000 datapoints. The results of our experiments are discussed in more detail in Section 6. Our conclusions and suggestions for future work are presented in Section 7.

数据传递给人类观察者。这极大地限制了这些技术在包含数千个高维数据点的现实数据集中的适用性。

与上述可视化技术不同，降维方法将高维数据集 $X = \{x_1, x_2, \dots, x_n\}$ 转换为二维或三维数据 $Y = \{y_1, y_2, \dots, y_n\}$ ，并可显示在散点图中。在论文中，我们将低维数据表示 Y 称为映射，将低维数据表示 y_i 单个数据点称为映射点。降维的目的是在低维映射中尽可能保留高维数据的重要结构。针对这一问题，已有多种技术被提出，这些技术在所保留的结构类型上有所不同。传统的降维技术，如主成分分析（PCA; 霍特林，1933 年）和经典多维尺度法（MDS; Torgerson, 1952）是一种线性技术，专注于保持不同数据点的低维表示之间保持距离。对于位于或接近低维非线性流形的高维数据，通常更重要的是将非常相似的数据点的低维表示保持得很近，而线性映射通常无法做到这一点。

大量旨在保持数据局部结构的非线性降维技术已被提出，其中许多由 Lee 和 Verleysen (2007) 综述。特别地，我们提到以下七种技术：(1) 萨蒙映射 (Sammon, 1969)，(2) 曲线分量分析 (CCA; Demartines 和 H'erault, 1997)，(3) 随机邻居嵌入 (SNE; Hinton 和 Roweis, 2002)，(4) 等位图 (Tenenbaum 等, 2000)，(5) 最大方差展开 (MVU; Weinberger 等, 2004)，(6) 局部线性嵌入 (LLE; Roweis 和 Saul, 2000)，以及 (7) 拉普拉斯特征图 (Belkin 和 Niyogi, 2002)。尽管这些技术在人工数据集上表现优异，但在可视化真实高维数据方面往往效果不佳。特别是，大多数技术无法在单一地图中同时保留数据的局部结构和全局结构。例如，一项最新研究显示，即使是半监督式的 MVU 变体，也无法将手写数字分离成其自然的群体 (Song 等, 2007)。

本文描述了将高维数据集转换为成对相似度矩阵的方法，并引入了一种称为“t-SNE”的新技术，用于可视化所得相似度数据。t-SNE 能够很好地捕提高维数据的局部结构，同时揭示多个尺度上的聚类等全局结构。我们通过将 t-SNE 与上述七种降维技术在多个领域中的五个数据集上进行比较，展示了其性能。由于篇幅限制，大部分 $(7+1) \times 5 = 40$ 图 $(7+1) \times 5 = 40$ 图谱已在补充材料中呈现，但论文中提供的图已足以证明 t-SNE 的优越性。

论文大纲如下。在第二部分，我们概述了 Hinton 和 Roweis (2002) 提出的 SNE，这构成了 t-SNE 的基础。在第三部分，我们介绍 t-SNE，它与 SNE 有两个重要区别。在第四部分，我们描述了实验装置和实验结果。随后，第 5 节展示了如何修改 t-SNE 以可视化包含超过 10,000 个数据点的现实世界数据集。我们的实验结果将在第 6 节中有更详细的讨论。我们的结论和未来工作的建议见第 7 节。

2. Stochastic Neighbor Embedding

Stochastic Neighbor Embedding (SNE) starts by converting the high-dimensional Euclidean distances between datapoints into conditional probabilities that represent similarities.¹ The similarity of datapoint x_j to datapoint x_i is the conditional probability, $p_{j|i}$, that x_i would pick x_j as its neighbor if neighbors were picked in proportion to their probability density under a Gaussian centered at x_i . For nearby datapoints, $p_{j|i}$ is relatively high, whereas for widely separated datapoints, $p_{j|i}$ will be almost infinitesimal (for reasonable values of the variance of the Gaussian, σ_i). Mathematically, the conditional probability $p_{j|i}$ is given by

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)}, \quad (1)$$

where σ_i is the variance of the Gaussian that is centered on datapoint x_i . The method for determining the value of σ_i is presented later in this section. Because we are only interested in modeling pairwise similarities, we set the value of $p_{i|i}$ to zero. For the low-dimensional counterparts y_i and y_j of the high-dimensional datapoints x_i and x_j , it is possible to compute a similar conditional probability, which we denote by $q_{j|i}$. We set² the variance of the Gaussian that is employed in the computation of the conditional probabilities $q_{j|i}$ to $\frac{1}{\sqrt{2}}$. Hence, we model the similarity of map point y_j to map point y_i by

$$q_{j|i} = \frac{\exp(-\|y_i - y_j\|^2)}{\sum_{k \neq i} \exp(-\|y_i - y_k\|^2)}.$$

Again, since we are only interested in modeling pairwise similarities, we set $q_{i|i} = 0$.

If the map points y_i and y_j correctly model the similarity between the high-dimensional datapoints x_i and x_j , the conditional probabilities $p_{j|i}$ and $q_{j|i}$ will be equal. Motivated by this observation, SNE aims to find a low-dimensional data representation that minimizes the mismatch between $p_{j|i}$ and $q_{j|i}$. A natural measure of the faithfulness with which $q_{j|i}$ models $p_{j|i}$ is the Kullback-Leibler divergence (which is in this case equal to the cross-entropy up to an additive constant). SNE minimizes the sum of Kullback-Leibler divergences over all datapoints using a gradient descent method. The cost function C is given by

$$C = \sum_i KL(P_i || Q_i) = \sum_i \sum_j p_{j|i} \log \frac{p_{j|i}}{q_{j|i}}, \quad (2)$$

in which P_i represents the conditional probability distribution over all other datapoints given datapoint x_i , and Q_i represents the conditional probability distribution over all other map points given map point y_i . Because the Kullback-Leibler divergence is not symmetric, different types of error in the pairwise distances in the low-dimensional map are not weighted equally. In particular, there is a large cost for using widely separated map points to represent nearby datapoints (i.e., for using

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1. SNE can also be applied to data sets that consist of pairwise similarities between objects rather than high-dimensional vector representations of each object, provided these similarities can be interpreted as conditional probabilities. For example, human word association data consists of the probability of producing each possible word in response to a given word, as a result of which it is already in the form required by SNE.
 2. Setting the variance in the low-dimensional Gaussians to another value only results in a rescaled version of the final map. Note that by using the same variance for every datapoint in the low-dimensional map, we lose the property that the data is a perfect model of itself if we embed it in a space of the same dimensionality, because in the high-dimensional space, we used a different variance σ_i in each Gaussian.

2. 随机邻居嵌入

随机邻域嵌入（SNE）首先将数据点之间的高维欧几里得距离转换为表示相似性的条件概率。数据点 x 与数据点 x 的相似度是指条件概率 p ，即如果邻居按以 x 为中心的高斯分布下概率密度比例被选中， x 会选择 x 作为邻居。对于相邻的数据点， p 相对较高，而对于距离较远的数据点， p 几乎是无穷小的（对于高斯方差的合理值， σ ）。数学上，条件概率 p 为

$$p = \frac{\exp\left(-\frac{\|x-x\|^2}{2\sigma^2}\right)}{\sum_{x'} \exp\left(-\frac{\|x-x'\|^2}{2\sigma^2}\right)}, \quad (1)$$

其中 σ 是以数据点 x 为中心的高斯方差。确定 σ 值的方法将在本节后面介绍。由于我们只关注两两相似性建模，因此将 p_{xx} 值设为零。对于高维数据点 x 和 x 的低维对应物 y 和 y ，可以计算出类似的条件概率，记为 q 。我们将用于条件概率 q 计算的高斯方差设为 σ_y 。因此，我们用 y （映射点 y ）来建模映射点 y （ y ）的相似性

$$q = \frac{\exp\left(-\frac{\|y-y\|^2}{2\sigma_y^2}\right)}{\sum_{y'} \exp\left(-\frac{\|y-y'\|^2}{2\sigma_y^2}\right)}.$$

同样，由于我们只关注两两相似度的建模，设 $q_{yy} = 0$ 。

如果映射点 y 和 y 正确建模了高维数据点 x 和 x 之间的相似性，条件概率 p 和 q 将相等。基于这一观察，SNE 旨在寻找一种低维数据表示，以最小化 p 与 q 之间的不匹配。 q 对 p 的忠实度的自然度量是 Kullback-Leibler 散度（此时等于交叉熵，且为加法常数）。SNE 利用梯度下降法最小化所有数据点上的 Kullback-Leibler 散度之和。成本函数 C 由下式给出

$$C = \sum_i \text{KL}(P_i \| Q_i) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}, \quad (2)$$

其中 P 表示给定数据点 x 的所有其他数据点的条件概率分布， Q 表示给定地图点 y 的所有其他映射点的条件概率分布。由于库尔巴克-莱布勒散度不对称，低维映射中两两距离中不同类型的误差权重不相等。特别地，使用相距甚远的映射点来表示附近数据点（即使用

1. SNE 也可以应用于由对象间成对相似性组成的数据集，而非每个对象的高维向量表示，前提是这些相似性可以被解释为条件概率。例如，人类词联想数据包含对给定词汇生成每个可能词汇的概率，因此该词已经是超声恩英语（SNE）要求的形式。2. 将低维高斯方差设置为另一个值，只会导致最终映射的重新缩放版本。注意，如果对低维映射中的每个数据点使用相同的方差，我们会失去数据作为自身完美模型的性质，因为在高维空间中，我们在每个高斯量中使用了不同的方差 σ 。

a small $q_{j|i}$ to model a large $p_{j|i}$), but there is only a small cost for using nearby map points to represent widely separated datapoints. This small cost comes from wasting some of the probability mass in the relevant Q distributions. In other words, the SNE cost function focuses on retaining the local structure of the data in the map (for reasonable values of the variance of the Gaussian in the high-dimensional space, σ_i).

The remaining parameter to be selected is the variance σ_i of the Gaussian that is centered over each high-dimensional datapoint, x_i . It is not likely that there is a single value of σ_i that is optimal for all datapoints in the data set because the density of the data is likely to vary. In dense regions, a smaller value of σ_i is usually more appropriate than in sparser regions. Any particular value of σ_i induces a probability distribution, P_i , over all of the other datapoints. This distribution has an entropy which increases as σ_i increases. SNE performs a binary search for the value of σ_i that produces a P_i with a fixed perplexity that is specified by the user.³ The perplexity is defined as

$$\text{Perp}(P_i) = 2^{H(P_i)},$$

where $H(P_i)$ is the Shannon entropy of P_i measured in bits

$$H(P_i) = - \sum_j p_{j|i} \log_2 p_{j|i}.$$

The perplexity can be interpreted as a smooth measure of the effective number of neighbors. The performance of SNE is fairly robust to changes in the perplexity, and typical values are between 5 and 50.

The minimization of the cost function in Equation 2 is performed using a gradient descent method. The gradient has a surprisingly simple form

$$\frac{\delta C}{\delta y_i} = 2 \sum_j (p_{j|i} - q_{j|i} + p_{i|j} - q_{i|j})(y_i - y_j).$$

Physically, the gradient may be interpreted as the resultant force created by a set of springs between the map point y_i and all other map points y_j . All springs exert a force along the direction $(y_i - y_j)$. The spring between y_i and y_j repels or attracts the map points depending on whether the distance between the two in the map is too small or too large to represent the similarities between the two high-dimensional datapoints. The force exerted by the spring between y_i and y_j is proportional to its length, and also proportional to its stiffness, which is the mismatch $(p_{j|i} - q_{j|i} + p_{i|j} - q_{i|j})$ between the pairwise similarities of the data points and the map points.

The gradient descent is initialized by sampling map points randomly from an isotropic Gaussian with small variance that is centered around the origin. In order to speed up the optimization and to avoid poor local minima, a relatively large momentum term is added to the gradient. In other words, the current gradient is added to an exponentially decaying sum of previous gradients in order to determine the changes in the coordinates of the map points at each iteration of the gradient search. Mathematically, the gradient update with a momentum term is given by

$$\mathcal{Y}^{(t)} = \mathcal{Y}^{(t-1)} + \eta \frac{\delta C}{\delta \mathcal{Y}} + \alpha(t) (\mathcal{Y}^{(t-1)} - \mathcal{Y}^{(t-2)}),$$

3. Note that the perplexity increases monotonically with the variance σ_i .

用小 Q 来建模大 p)，但使用附近的映射点表示相距较远的数据点成本很小。这种小成本来自于在相关 Q 分布中浪费了部分概率质量。换句话说，SNE 成本函数侧重于保持映射中数据的局部结构（对于高维空间中高斯方差的合理值，为 σ ）。

剩余的参数是高斯分布方差 σ ，该高斯分布中心于每个高维数据点 x 。由于数据密度可能变化，不太可能存在对数据集中所有数据点都最优的 σ 值。在密集区域，较小的 σ 通常比稀疏区域更合适。任何特定的 σ 值都会诱导出一个概率分布 P ，覆盖所有其他数据点。该分布的熵随 σ 增加而增加。SNE 对 σ 进行二分搜索，生成一个由用户指定的固定困惑度的 P 。困惑度定义为

$$\text{Perp}(P) = 2,$$

其中 $H(P)$ 是 P 的香农熵，单位为比特

$$H(P) = -\sum_j p_j \log p_j.$$

困惑度可以被理解为有效邻居数量的平滑度量。SNE 对困惑度变化表现相当稳健，典型值在 5 到 50 之间。

方程 2 中成本函数的最小化是通过梯度下降法实现的。梯度形式出奇地简单

$$\frac{\delta C}{\delta y} = 2 \sum (p - q + p - q) (y - y).$$

物理上，梯度可解释为由一组弹簧在映射点 y 与所有其他映射点 y 之间产生的合力。所有弹簧沿方向 $(y - y)$ 施加力。 y 和 y 之间的弹簧会排斥或吸引映射点，取决于两者之间的距离是过大还是太短，无法表示两个高维数据点之间的相似性。弹簧在 y 和 y 之间施加的力与其长度成正比，也与刚度成正比，刚度即数据点与映射点两两相似度之间的不匹配 $(p - q + p - q)$ 。

梯度下降通过随机抽样从一个方差较小的同性高斯图中随机抽样，该高斯图以原点为中心来初始化。为了加快优化速度并避免局部极小值不佳，梯度中加入一个较大的动量项。换句话说，当前梯度被加到之前梯度的指数衰减和中，以确定每次梯度搜索中地图点坐标的变化。数学上，带动量项的梯度更新为

$$y^{(t)} = y + \eta \left(\frac{\delta C}{\delta y} + \alpha (y^{(t-1)} - y) \right),$$

3. 注意，困惑度随方差 σ 单调增加。

where $\mathcal{Y}^{(t)}$ indicates the solution at iteration t , η indicates the learning rate, and $\alpha(t)$ represents the momentum at iteration t .

In addition, in the early stages of the optimization, Gaussian noise is added to the map points after each iteration. Gradually reducing the variance of this noise performs a type of simulated annealing that helps the optimization to escape from poor local minima in the cost function. If the variance of the noise changes very slowly at the critical point at which the global structure of the map starts to form, SNE tends to find maps with a better global organization. Unfortunately, this requires sensible choices of the initial amount of Gaussian noise and the rate at which it decays. Moreover, these choices interact with the amount of momentum and the step size that are employed in the gradient descent. It is therefore common to run the optimization several times on a data set to find appropriate values for the parameters.⁴ In this respect, SNE is inferior to methods that allow convex optimization and it would be useful to find an optimization method that gives good results without requiring the extra computation time and parameter choices introduced by the simulated annealing.

3. t-Distributed Stochastic Neighbor Embedding

Section 2 discussed SNE as it was presented by Hinton and Roweis (2002). Although SNE constructs reasonably good visualizations, it is hampered by a cost function that is difficult to optimize and by a problem we refer to as the “crowding problem”. In this section, we present a new technique called “t-Distributed Stochastic Neighbor Embedding” or “t-SNE” that aims to alleviate these problems. The cost function used by t-SNE differs from the one used by SNE in two ways: (1) it uses a symmetrized version of the SNE cost function with simpler gradients that was briefly introduced by Cook et al. (2007) and (2) it uses a Student-t distribution rather than a Gaussian to compute the similarity between two points *in the low-dimensional space*. t-SNE employs a heavy-tailed distribution in the low-dimensional space to alleviate both the crowding problem and the optimization problems of SNE.

In this section, we first discuss the symmetric version of SNE (Section 3.1). Subsequently, we discuss the crowding problem (Section 3.2), and the use of heavy-tailed distributions to address this problem (Section 3.3). We conclude the section by describing our approach to the optimization of the t-SNE cost function (Section 3.4).

3.1 Symmetric SNE

As an alternative to minimizing the sum of the Kullback-Leibler divergences between the conditional probabilities $p_{j|i}$ and $q_{j|i}$, it is also possible to minimize a single Kullback-Leibler divergence between a joint probability distribution, P , in the high-dimensional space and a joint probability distribution, Q , in the low-dimensional space:

$$C = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}.$$

where again, we set p_{ii} and q_{ii} to zero. We refer to this type of SNE as symmetric SNE, because it has the property that $p_{ij} = p_{ji}$ and $q_{ij} = q_{ji}$ for $\forall i, j$. In symmetric SNE, the pairwise similarities in

4. Picking the best map after several runs as a visualization of the data is not nearly as problematic as picking the model that does best on a test set during supervised learning. In visualization, the aim is to see the structure in the training data, not to generalize to held out test data.

表示迭代 t 时的解， η 表示学习速率， $\alpha(t)$ 表示迭代 t 时的动量。

此外，在优化的早期阶段，每次迭代后都会向映射点添加高斯噪声。逐渐降低噪声的方差，可以进行一种模拟退火，帮助优化摆脱成本函数的局部极小值。如果噪声的方差在地图全局结构开始形成的临界点变化非常缓慢，SNE 倾向于找到具有更好全局组织的映射。不幸的是，这需要合理地选择初始高斯噪声量及其衰减速率。此外，这些选择还与梯度下降中使用的动量和步长相互作用。因此，通常会对数据集多次运行优化以寻找合适的参数值。在这方面，SNE 不如允许凸优化的方法，找到一种既能带来良好结果又不需模拟退火带来额外计算时间和参数选择的优化方法将非常有用。

3. t-分布随机邻居嵌入

第二节讨论了 Hinton 和 Roweis (2002) 提出的 SNE。尽管 SNE 构建了相当不错的可视化，但它受到一个难以优化的成本函数以及我们称之为“拥挤问题”的问题的阻碍。在本节中，我们介绍一种名为“t-分布随机邻居嵌入”(t-SNE)“的新技术，旨在缓解这些问题。t-SNE 所用的成本函数与 SNE 的有两个不同点：(1) 它使用了 Cook 等人 (2007) 简要引入的对称化版本、梯度更简单；(2) 它使用学生-t 分布而非高斯分布来计算低维空间中两点的相似性。t-SNE 在低维空间中采用重尾分布，以缓解 SNE 的拥挤问题和优化问题。

本节首先讨论 SNE 的对称版本 (第 3.1 节)。随后，我们将讨论拥挤问题 (第 3.2 节)，以及利用重尾分布解决这一问题的方法 (第 3.3 节)。我们以描述优化 t-SNE 成本函数的方法作为本节结尾 (第 3.4 节)。

3.1 对称 SNE

作为最小化条件概率 p 和 q 之间 Kullback-Leibler 散度和的替代方案，也可以在高维空间中最小化联合概率分布 P 与低维空间中联合概率分布 Q 之间的单一 Kullback-Leibler 散度：

$$C = KL(P||Q) = \sum_i \sum_j \frac{p_{ij}}{q_{ij}}.$$

同样，我们将 p_{ii} 和 q_{ii} 设为零。我们称这种类型的 SNE 为对称 SNE，因为它具有 $\forall i, j$ 时 $p_{ij} = p_{ji}$ 和 $q_{ij} = q_{ji}$ 的性质。在对称 SNE 中，两两相似性

4. 在多次运行后选择最佳地图作为数据可视化，远不如在监督学习中选择测试集表现最佳的模型那么棘手。在可视化中，目标是看到训练数据中的结构，而不是推广到保留的测试数据。

the low-dimensional map q_{ij} are given by

$$q_{ij} = \frac{\exp(-\|y_i - y_j\|^2)}{\sum_{k \neq l} \exp(-\|y_k - y_l\|^2)}, \quad (3)$$

The obvious way to define the pairwise similarities in the high-dimensional space p_{ij} is

$$p_{ij} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma^2)}{\sum_{k \neq l} \exp(-\|x_k - x_l\|^2 / 2\sigma^2)},$$

but this causes problems when a high-dimensional datapoint x_i is an outlier (i.e., all pairwise distances $\|x_i - x_j\|^2$ are large for x_i). For such an outlier, the values of p_{ij} are extremely small for all j , so the location of its low-dimensional map point y_i has very little effect on the cost function. As a result, the position of the map point is not well determined by the positions of the other map points. We circumvent this problem by defining the joint probabilities p_{ij} in the high-dimensional space to be the symmetrized conditional probabilities, that is, we set $p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$. This ensures that $\sum_j p_{ij} > \frac{1}{2n}$ for all datapoints x_i , as a result of which each datapoint x_i makes a significant contribution to the cost function. In the low-dimensional space, symmetric SNE simply uses Equation 3. The main advantage of the symmetric version of SNE is the simpler form of its gradient, which is faster to compute. The gradient of symmetric SNE is fairly similar to that of asymmetric SNE, and is given by

$$\frac{\delta C}{\delta y_i} = 4 \sum_j (p_{ij} - q_{ij})(y_i - y_j).$$

In preliminary experiments, we observed that symmetric SNE seems to produce maps that are just as good as asymmetric SNE, and sometimes even a little better.

3.2 The Crowding Problem

Consider a set of datapoints that lie on a two-dimensional curved manifold which is approximately linear on a small scale, and which is embedded within a higher-dimensional space. It is possible to model the small pairwise distances between datapoints fairly well in a two-dimensional map, which is often illustrated on toy examples such as the “Swiss roll” data set. Now suppose that the manifold has ten intrinsic dimensions⁵ and is embedded within a space of much higher dimensionality. There are several reasons why the pairwise distances in a two-dimensional map cannot faithfully model distances between points on the ten-dimensional manifold. For instance, in ten dimensions, it is possible to have 11 datapoints that are mutually equidistant and there is no way to model this faithfully in a two-dimensional map. A related problem is the very different distribution of pairwise distances in the two spaces. The volume of a sphere centered on datapoint i scales as r^m , where r is the radius and m the dimensionality of the sphere. So if the datapoints are approximately uniformly distributed in the region around i on the ten-dimensional manifold, and we try to model the distances from i to the other datapoints in the two-dimensional map, we get the following “crowding problem”: the area of the two-dimensional map that is available to accommodate moderately distant datapoints will not be nearly large enough compared with the area available to accommodate nearby datapoints. Hence, if we want to model the small distances accurately in the map, most of the points

5. This is approximately correct for the images of handwritten digits we use in our experiments in Section 4.

低维映射 q 由下式给出

$$q = \frac{\exp(-\|y - y\|)}{\sum \exp(-\|y - y\|)}, \quad (3)$$

定义高维空间 p 中两两相似性的显而易见的方法

$$p = \frac{\exp(-\|x - x\|/2\sigma)}{\sum \exp(-\|x - x\|/2\sigma)},$$

但当高维数据点 x 是离群值（即所有两两距离 $\|x - x\|$ 对 x 都很大）时，这会带来问题。对于这样的离群值， p 值对所有 j 都极小，因此其低维映射点 y 的位置对代价函数影响极小。因此，地图点的位置无法由其他地图点的位置很好地确定。我们通过将高维空间中的联合概率定义为对称化条件概率来规避此问题，即设 $p =$

$\frac{p_i + p_j}{2}$ 。这确保了 $\sum$ 对所有数据点 x 的 $p > 0$ ，因此每个数据点 x 对成本函数贡献显著。在低维空间中，对称 SNE 简单地使用方程 3。对称版本 SNE 的主要优点是梯度形式更简单，计算更快。对称 SNE 的梯度与非对称 SNE 的梯度相当相似，给出为

$$\frac{\delta C}{\delta y} = 4 \sum (p - q) (y - y)。$$

在初步实验中，我们观察到对称 SNE 产生的映射质量与非对称 SNE 相当，有时甚至更好。

3.2 拥挤问题

考虑一组位于二维曲面流形上的数据点，该流形在小尺度上近似线性，且嵌入在高维空间中。在二维地图中，可以较好地模拟数据点之间的小两两距离，这通常在“瑞士卷”数据集等玩具示例中有所体现。现在假设流形有十个内在维度，嵌入在一个维度更高的空间中。二维地图中的两两距离无法忠实模拟十维流形上点之间的距离有几个原因。例如，在十维空间中，可能有 11 个相互等距的数据点，而在二维地图上无法忠实建模。一个相关的问题是空间中两两距离分布的巨大差异。以数据点 i 为中心的球体体积随 r 比例缩放，其中 r 为半径， m 为球体的维度。因此，如果数据点大致均匀分布在十维流形中 i 周围区域，并试图建模 i 到二维地图中其他数据点的距离，就会得到以下“拥挤问题”：二维地图中可用容纳中等距离数据点的面积远远不够大，而可用面积则远远不足以容纳附近数据点。因此，如果我们想准确模拟地图上的小距离，大多数点

5. 这大致符合我们在第 4 节实验中使用的手写数字图像。

that are at a moderate distance from datapoint i will have to be placed much too far away in the two-dimensional map. In SNE, the spring connecting datapoint i to each of these too-distant map points will thus exert a very small attractive force. Although these attractive forces are very small, the very large number of such forces crushes together the points in the center of the map, which prevents gaps from forming between the natural clusters. Note that the crowding problem is not specific to SNE, but that it also occurs in other local techniques for multidimensional scaling such as Sammon mapping.

An attempt to address the crowding problem by adding a slight repulsion to all springs was presented by Cook et al. (2007). The slight repulsion is created by introducing a uniform background model with a small mixing proportion, ρ . So however far apart two map points are, q_{ij} can never fall below $\frac{2\rho}{n(n-1)}$ (because the uniform background distribution is over $n(n-1)/2$ pairs). As a result, for datapoints that are far apart in the high-dimensional space, q_{ij} will always be larger than p_{ij} , leading to a slight repulsion. This technique is called UNI-SNE and although it usually outperforms standard SNE, the optimization of the UNI-SNE cost function is tedious. The best optimization method known is to start by setting the background mixing proportion to zero (i.e., by performing standard SNE). Once the SNE cost function has been optimized using simulated annealing, the background mixing proportion can be increased to allow some gaps to form between natural clusters as shown by Cook et al. (2007). Optimizing the UNI-SNE cost function directly does not work because two map points that are far apart will get almost all of their q_{ij} from the uniform background. So even if their p_{ij} is large, there will be no attractive force between them, because a small change in their separation will have a vanishingly small *proportional* effect on q_{ij} . This means that if two parts of a cluster get separated early on in the optimization, there is no force to pull them back together.

3.3 Mismatched Tails can Compensate for Mismatched Dimensionalities

Since symmetric SNE is actually matching the joint probabilities of pairs of datapoints in the high-dimensional and the low-dimensional spaces rather than their distances, we have a natural way of alleviating the crowding problem that works as follows. In the high-dimensional space, we convert distances into probabilities using a Gaussian distribution. In the low-dimensional map, we can use a probability distribution that has much heavier tails than a Gaussian to convert distances into probabilities. This allows a moderate distance in the high-dimensional space to be faithfully modeled by a much larger distance in the map and, as a result, it eliminates the unwanted attractive forces between map points that represent moderately dissimilar datapoints.

In t-SNE, we employ a Student t-distribution with one degree of freedom (which is the same as a Cauchy distribution) as the heavy-tailed distribution in the low-dimensional map. Using this distribution, the joint probabilities q_{ij} are defined as

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}. \quad (4)$$

We use a Student t-distribution with a single degree of freedom, because it has the particularly nice property that $(1 + \|y_i - y_j\|^2)^{-1}$ approaches an inverse square law for large pairwise distances $\|y_i - y_j\|$ in the low-dimensional map. This makes the map's representation of joint probabilities (almost) invariant to changes in the scale of the map for map points that are far apart. It also means that large clusters of points that are far apart interact in just the same way as individual points, so the optimization operates in the same way at all but the finest scales. A theoretical justification for our

距离数据点 i 中等距离的，在二维地图上必须被放置在过远的位置。在特殊偏向空间中，连接数据点 i 与这些过远地图点的弹簧因此会产生非常小的吸引力。虽然这些吸引力非常小，但大量这样的力会将地图中心的点挤压在一起，从而防止自然星团之间形成间隙。需要注意的是，拥挤问题并非 SNE 特有，但它也存在于其他多维尺度的局部技术中，如萨蒙映射。

Cook 等人 (2007) 提出了通过在所有弹簧上增加轻微排斥力来解决拥挤问题的尝试。轻微排斥是通过引入一个均匀背景模型，混合比例小 ρ 来实现的。因此，无论两个映射点相距多远， q 都不可能低于 n (因为均匀背景分布分布为 $n(n-1)/2$ 对)。因此，对于高维空间中相距甚远的数据点， q 总是大于 p ，导致轻微排斥。该技术称为 UNI-SNE，虽然通常优于标准 SNE，但 UNI-SNE 成本函数的优化过程较为繁琐。已知的最佳优化方法是先将背景混合比例设为零 (即执行标准单一电子平均。一旦通过模拟退火优化了 SNE 成本函数，背景混合比例可以增加，以便自然簇间形成一些间隙，正如 Cook 等人 (2007) 所示。直接优化 UNI-SNE 成本函数行不通，因为两个相距较远的映射点几乎所有 q 都来自均匀背景。所以即使它们的 π 很大，它们之间也不会有吸引力，因为它们分离的一点点变化对 q 的比例效应极其微小。这意味着如果集群的两个部分在优化早期被分离，就没有强制力将它们拉回。

3.3 不匹配的尾巴可以补偿不匹配的维度

由于对称 SNE 实际上匹配的是高维和低维空间中数据点对的联合概率，而非它们的距离，我们有一种自然的方法来缓解拥挤问题，具体工作原理如下。在高维空间中，我们利用高斯分布将距离转换为概率。在低维映射中，我们可以使用比高斯分布尾部更重的概率分布，将距离转换为概率。这使得高维空间中中适中的距离可以用地图中更大的距离忠实建模，从而消除代表中等差异数据点的地图点之间不必要的吸引力。

在 t-SNE 中，我们采用了具有一自由度的学生 t 分布 (与柯西分布相同)，与低维映射中的重尾分布相同。利用该分布，联合概率 q_{ij} 定义为

$$q_{ij} = \frac{\left(1 + \frac{\|y_i - y_j\|}{2}\right)^{-5}}{\sum_k \left(1 + \frac{\|y_i - y_k\|}{2}\right)^{-5}}. \quad (4)$$

我们使用具有单自由度的学生 t 分布，因为它具有一个特别不错的性质

对于低维映射中较大的成对距离 $\|y_i - y_j\|$ ，趋近于反平方定律。这使得地图对联合概率的表示 (几乎) 对距离较远的地图点尺度变化保持不变。这也意味着距离较远的点群以相同的方式相互作用，因此优化在最细微的尺度上几乎以相同方式运作。对我们

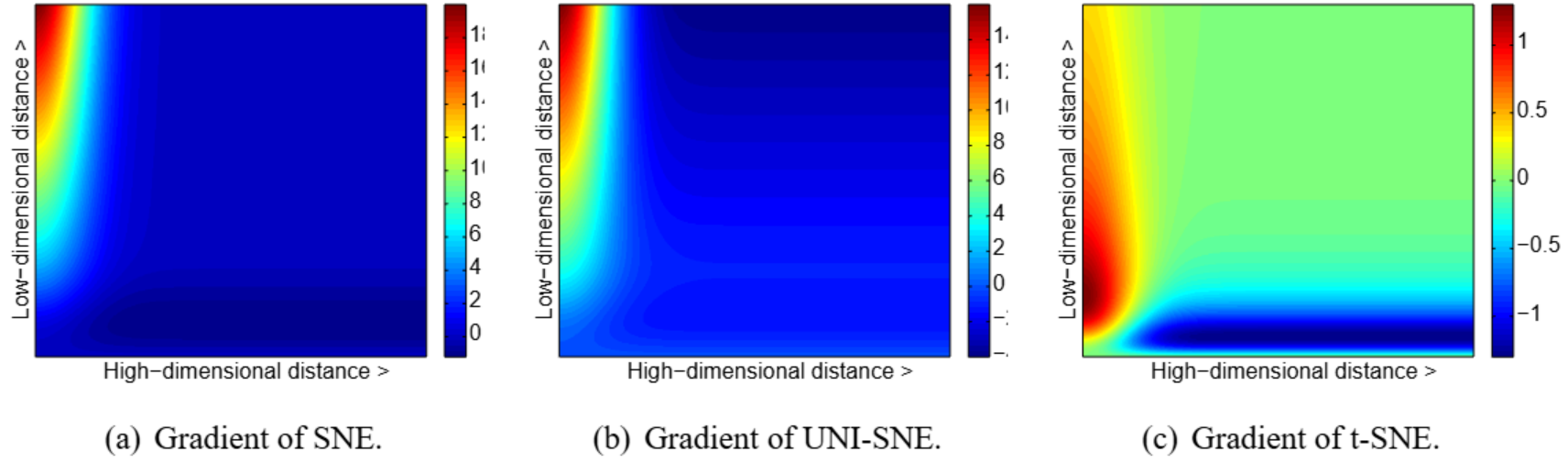


Figure 1: Gradients of three types of SNE as a function of the pairwise Euclidean distance between two points in the high-dimensional and the pairwise distance between the points in the low-dimensional data representation.

selection of the Student t-distribution is that it is closely related to the Gaussian distribution, as the Student t-distribution is an infinite mixture of Gaussians. A computationally convenient property is that it is much faster to evaluate the density of a point under a Student t-distribution than under a Gaussian because it does not involve an exponential, even though the Student t-distribution is equivalent to an infinite mixture of Gaussians with different variances.

The gradient of the Kullback-Leibler divergence between P and the Student-t based joint probability distribution Q (computed using Equation 4) is derived in Appendix A, and is given by

$$\frac{\delta C}{\delta y_i} = 4 \sum_j (p_{ij} - q_{ij}) (y_i - y_j) (1 + \|y_i - y_j\|^2)^{-1}. \quad (5)$$

In Figure 1(a) to 1(c), we show the gradients between two low-dimensional datapoints y_i and y_j as a function of their pairwise Euclidean distances in the high-dimensional and the low-dimensional space (i.e., as a function of $\|x_i - x_j\|$ and $\|y_i - y_j\|$) for the symmetric versions of SNE, UNI-SNE, and t-SNE. In the figures, positive values of the gradient represent an attraction between the low-dimensional datapoints y_i and y_j , whereas negative values represent a repulsion between the two datapoints. From the figures, we observe two main advantages of the t-SNE gradient over the gradients of SNE and UNI-SNE.

First, the t-SNE gradient strongly repels dissimilar datapoints that are modeled by a small pairwise distance in the low-dimensional representation. SNE has such a repulsion as well, but its effect is minimal compared to the strong attractions elsewhere in the gradient (the largest attraction in our graphical representation of the gradient is approximately 19, whereas the largest repulsion is approximately 1). In UNI-SNE, the amount of repulsion between dissimilar datapoints is slightly larger, however, this repulsion is only strong when the pairwise distance between the points in the low-dimensional representation is already large (which is often not the case, since the low-dimensional representation is initialized by sampling from a Gaussian with a very small variance that is centered around the origin).

Second, although t-SNE introduces strong repulsions between dissimilar datapoints that are modeled by small pairwise distances, these repulsions do not go to infinity. In this respect, t-SNE differs from UNI-SNE, in which the strength of the repulsion between very dissimilar datapoints

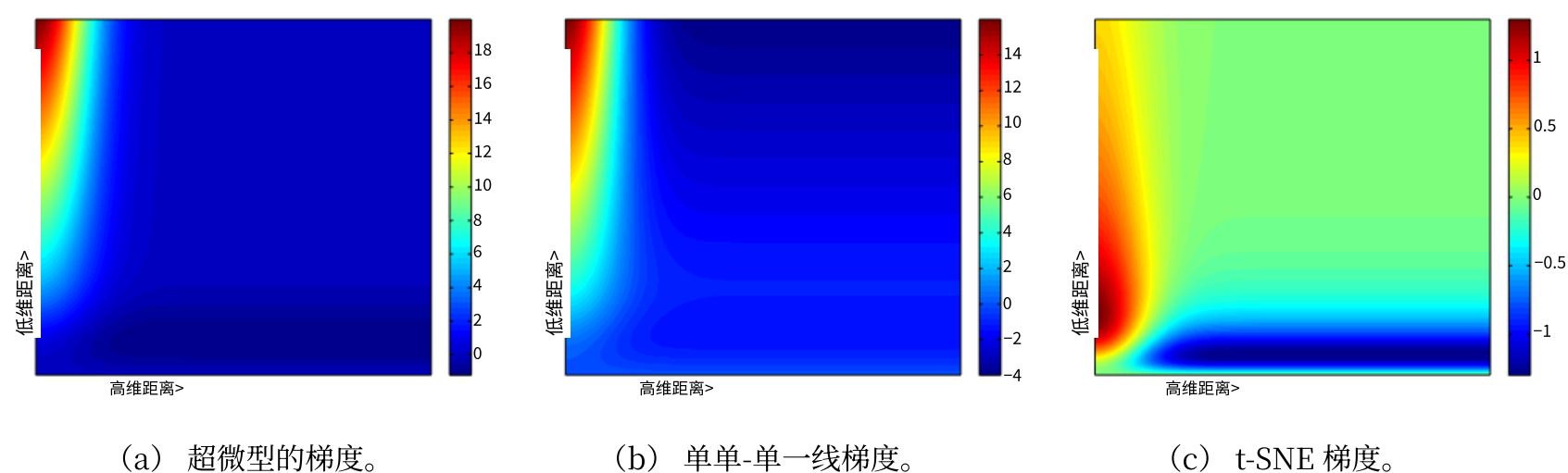


图 1: 三种类型 SNE 作为两两欧几里得距离函数的梯度
高维中的两个点，以及低维数据表示中点之间的两对距离。

学生 t 分布的选择在于它与高斯分布密切相关，因为学生 t 分布是高斯分布的无限混合。计算上一个方便的性质是，在学生 t 分布下评估点的密度比在高斯分布下快得多，因为它不涉及指数分布，尽管学生 t 分布等价于具有不同方差的高斯分布的无限混合。

P 与基于学生 t 的联合概率分布 Q（使用方程 4 计算）之间 Kullback-Leibler 散度的梯度见附录 A，给出为

$$\frac{\delta C}{\delta y} = 4 \sum (p - q) (y - y) (1 + \|y - y\|) . \quad (5)$$

在图 1 (a) 到 1 (c) 中，我们展示了两个低维数据点 y 和 y 之间的梯度，作为它们在高维和低维空间中两两欧几里得距离的函数（即对称的 SNE、单 SNE 和 t-SNE 的函数，分别为 $\|x - x\|$ 和 $\|y - y\|$ ）。图中，梯度的正值表示低维数据点 y 和 y 之间的吸引力，负值表示两个数据点之间的排斥。从图中，我们观察到 t-SNE 梯度相较于 SNE 和 UNI-SNE 梯度的两个主要优势。

首先，t-SNE 梯度强烈排斥以低维表示中小成对距离建模的异类数据点。SNE 也有这种排斥力，但其影响相较于梯度中其他强吸引（我们图形化的梯度最大吸引约为 19，最大排斥约为 1）。在 UNI-SNE 中，不同数据点之间的排斥量略大，但这种排斥只有在低维表示中点之间的两对距离已经很大时才会强（通常情况下并非如此，因为低维表示是从以原点为中心的高斯方差极小的抽样中初始化的）。

其次，尽管 t-SNE 引入了不同数据点之间的强烈排斥，这些数据点由小的两两距离建模，但这些排斥不会无限远。在这方面，t-SNE 与 UNI-SNE 不同，后者在非常不同数据点之间的排斥强度不同

Algorithm 1: Simple version of t-Distributed Stochastic Neighbor Embedding.

Data: data set $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$,
cost function parameters: perplexity $Perp$,
optimization parameters: number of iterations T , learning rate η , momentum $\alpha(t)$.
Result: low-dimensional data representation $\mathcal{Y}^{(T)} = \{y_1, y_2, \dots, y_n\}$.

begin
 compute pairwise affinities $p_{j|i}$ with perplexity $Perp$ (using Equation 1)
 set $p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$
 sample initial solution $\mathcal{Y}^{(0)} = \{y_1, y_2, \dots, y_n\}$ from $\mathcal{N}(0, 10^{-4}I)$
 for $t=1$ **to** T **do**
 compute low-dimensional affinities q_{ij} (using Equation 4)
 compute gradient $\frac{\delta \mathcal{C}}{\delta \mathcal{Y}}$ (using Equation 5)
 set $\mathcal{Y}^{(t)} = \mathcal{Y}^{(t-1)} + \eta \frac{\delta \mathcal{C}}{\delta \mathcal{Y}} + \alpha(t) (\mathcal{Y}^{(t-1)} - \mathcal{Y}^{(t-2)})$
 end
end

is proportional to their pairwise distance in the low-dimensional map, which may cause dissimilar datapoints to move much too far away from each other.

Taken together, t-SNE puts emphasis on (1) modeling dissimilar datapoints by means of large pairwise distances, and (2) modeling similar datapoints by means of small pairwise distances. Moreover, as a result of these characteristics of the t-SNE cost function (and as a result of the approximate scale invariance of the Student t-distribution), the optimization of the t-SNE cost function is much easier than the optimization of the cost functions of SNE and UNI-SNE. Specifically, t-SNE introduces long-range forces in the low-dimensional map that can pull back together two (clusters of) similar points that get separated early on in the optimization. SNE and UNI-SNE do not have such long-range forces, as a result of which SNE and UNI-SNE need to use simulated annealing to obtain reasonable solutions. Instead, the long-range forces in t-SNE facilitate the identification of good local optima without resorting to simulated annealing.

3.4 Optimization Methods for t-SNE

We start by presenting a relatively simple, gradient descent procedure for optimizing the t-SNE cost function. This simple procedure uses a momentum term to reduce the number of iterations required and it works best if the momentum term is small until the map points have become moderately well organized. Pseudocode for this simple algorithm is presented in Algorithm 1. The simple algorithm can be sped up using the adaptive learning rate scheme that is described by Jacobs (1988), which gradually increases the learning rate in directions in which the gradient is stable.

Although the simple algorithm produces visualizations that are often much better than those produced by other non-parametric dimensionality reduction techniques, the results can be improved further by using either of two tricks. The first trick, which we call “early compression”, is to force the map points to stay close together at the start of the optimization. When the distances between map points are small, it is easy for clusters to move through one another so it is much easier to explore the space of possible global organizations of the data. Early compression is implemented by adding an additional L2-penalty to the cost function that is proportional to the sum of squared

算法 1: t 分布随机邻居嵌入的简单版本。	
数据: 数据集 $X = \{x_1, x_2, \dots, x_n\}$, 成本函数参数: 困惑度 Perp, 优化参数: 迭代次数 T , 学习率 η , 动量 $\alpha(t)$ 。	
结果: 低维数据表示	$Y^{(T)} = \{y_1, y_2, \dots, y_n\}$ 。
开始	
计算两对亲和力和 p , Perp (使用方程 1) 集合 $p = \frac{p_{ij} + p_{ji}}{2n}$	
从 $N(0, 10I)$ 中取样初始解 $Y = \{y_1, y_2, \dots, y_n\}$	
对于 $t=1$ 到 T , 则	
计算低维亲和力和 q (使用方程 4), 计算梯度 Y (使用方程 5), 集合 $Y = Y + \eta \nabla Y + \alpha(t) (Y^{(t-1)} - Y)$	
结束	
结束	

与它们在低维地图上的两对距离成正比，这可能导致不同数据点之间移动得过远。综合来看，t-SNE 强调：(1) 通过较大的成对距离建模不同数据点，以及 (2) 通过较小的成对距离来建模相似数据点。此外，由于 t-SNE 成本函数的这些特性（以及学生 t 分布的近似尺度不变性），t-SNE 成本函数的优化比 SNE 和 UNI-SNE 的成本函数优化要容易得多。具体来说，t-SNE 在低维映射中引入了长程力，可以将两个（由）相似点组成的簇重新聚合，这些点在优化初期就被分离。SNE 和 UNI-SNE 没有这样的远程力，因此 SNE 和 UNI-SNE 需要使用模拟退火来获得合理的解。相反，t-超核电子中的长程力有助于识别良好的局部最优解，而无需模拟退火。

3.4 t-SNE 优化方法

我们首先介绍一个相对简单的梯度下降过程，用于优化 t-SNE 成本函数。该简单过程利用动量项减少迭代次数，且在动量项较小、映射点组织较为完善之前效果最佳。该简单算法的伪代码见算法 1。该简单算法可以通过 Jacobs (1988) 描述的自适应学习率方案加速，该方案在梯度稳定的方向上逐步提高学习率。虽然该简单算法产生的可视化效果通常远优于其他非参数降维技术，但通过使用两种技巧中的任一，结果仍可进一步改进。第一个技巧，我们称之为“早期压缩”，是强制映射点在优化开始时保持彼此靠近。当地图点之间的距离较小时，簇之间容易相互移动，因此探索数据可能的全球组织空间会更加容易。早期压缩通过在成本函数中增加一个与平方和成正比的额外 L2-惩罚来实现

distances of the map points from the origin. The magnitude of this penalty term and the iteration at which it is removed are set by hand, but the behavior is fairly robust across variations in these two additional optimization parameters.

A less obvious way to improve the optimization, which we call “early exaggeration”, is to multiply all of the p_{ij} ’s by, for example, 4, in the initial stages of the optimization. This means that almost all of the q_{ij} ’s, which still add up to 1, are much too small to model their corresponding p_{ij} ’s. As a result, the optimization is encouraged to focus on modeling the large p_{ij} ’s by fairly large q_{ij} ’s. The effect is that the natural clusters in the data tend to form tight widely separated clusters in the map. This creates a lot of relatively empty space in the map, which makes it much easier for the clusters to move around relative to one another in order to find a good global organization.

In all the visualizations presented in this paper and in the supporting material, we used exactly the same optimization procedure. We used the early exaggeration method with an exaggeration of 4 for the first 50 iterations (note that early exaggeration is not included in the pseudocode in Algorithm 1). The number of gradient descent iterations T was set 1000, and the momentum term was set to $\alpha^{(t)} = 0.5$ for $t < 250$ and $\alpha^{(t)} = 0.8$ for $t \geq 250$. The learning rate η is initially set to 100 and it is updated after every iteration by means of the adaptive learning rate scheme described by Jacobs (1988). A Matlab implementation of the resulting algorithm is available at <http://ticc.uvt.nl/~lvdrmaaten/tsne>.

4. Experiments

To evaluate t-SNE, we present experiments in which t-SNE is compared to seven other non-parametric techniques for dimensionality reduction. Because of space limitations, in the paper, we only compare t-SNE with: (1) Sammon mapping, (2) Isomap, and (3) LLE. In the supporting material, we also compare t-SNE with: (4) CCA, (5) SNE, (6) MVU, and (7) Laplacian Eigenmaps. We performed experiments on five data sets that represent a variety of application domains. Again because of space limitations, we restrict ourselves to three data sets in the paper. The results of our experiments on the remaining two data sets are presented in the supplemental material.

In Section 4.1, the data sets that we employed in our experiments are introduced. The setup of the experiments is presented in Section 4.2. In Section 4.3, we present the results of our experiments.

4.1 Data Sets

The five data sets we employed in our experiments are: (1) the MNIST data set, (2) the Olivetti faces data set, (3) the COIL-20 data set, (4) the word-features data set, and (5) the Netflix data set. We only present results on the first three data sets in this section. The results on the remaining two data sets are presented in the supporting material. The first three data sets are introduced below.

The MNIST data set⁶ contains 60,000 grayscale images of handwritten digits. For our experiments, we randomly selected 6,000 of the images for computational reasons. The digit images have $28 \times 28 = 784$ pixels (i.e., dimensions). The Olivetti faces data set⁷ consists of images of 40 individuals with small variations in viewpoint, large variations in expression, and occasional addition of glasses. The data set consists of 400 images (10 per individual) of size $92 \times 112 = 10,304$ pixels, and is labeled according to identity. The COIL-20 data set (Nene et al., 1996) contains images of 20

6. The MNIST data set is publicly available from <http://yann.lecun.com/exdb/mnist/index.html>.

7. The Olivetti faces data set is publicly available from <http://mambo.ucsc.edu/psl/olivetti.html>.

地图指向原点的距离。该惩罚项的大小及其去除迭代由人工设定，但该行为在这两个额外优化参数的变化中相当稳健。

一种不那么明显的优化方法，我们称之为“早期夸张”，是在优化初期将所有 p 乘以例如 4。这意味着几乎所有的 q （加起来仍为 1）都太小，无法模拟对应的 p 。因此，优化建议将较大的 p 模型化为较大的 q 。结果是数据中的自然聚类往往在地图上形成紧密且分散的聚类。这会在地图上产生大量相对空白的空间，使集群之间相对移动变得更容易，从而找到一个良好的全球组织。

在本文及相关资料中展示的所有可视化中，我们使用了完全相同的优化过程。我们使用了早期夸张方法，前 50 次迭代夸张为 4（注意算法 1 的伪代码中未包含早期夸张）。梯度下降迭代次数 T 设为 1000，动量项设为 $\alpha = 0.5$ ， $t < 250$ 时 $\alpha = 0.8$ ， $t \geq 250$ 。学习率 η 初始设置为 100，每次迭代后通过 Jacobs (1988) 描述的自适应学习率方案进行更新。该算法的 Matlab 实现可在 <http://ticc> 提供。

uvt.nl/~lvdrmaaten/tsne.

4. 实验

为评估 t-SNE，我们提出实验，将 t-SNE 与其他七种非参数降维技术进行比较。由于篇幅限制，本文仅将 t-SNE 与以下对象进行比较：(1) 萨蒙映射，(2) 同位图，(3) LLE。在辅助材料中，我们还比较了 t-SNE 与：(4) CCA，(5) SNE，(6) MVU，以及 (7) 拉普拉斯特征图。我们对五个代表不同应用领域的数据集进行了实验。同样由于篇幅限制，我们在论文中限制了三个数据集。我们对剩余两个数据集的实验结果收录于补充材料中。

在第 4.1 节，介绍了我们在实验中使用的数据集。实验的设置见第 4.2 节。在第 4.3 节，我们展示了实验结果。

4.1 数据集

我们在实验中使用的五个数据集分别是：(1) MNIST 数据集，(2) Olivetti Faces 数据集，(3) COIL-20 数据集，(4) 单词特征数据集，以及 (5) Netflix 数据集。本节仅呈现前三个数据集的结果。剩余两个数据集的结果见支持材料。以下介绍前三个数据集。

MNIST 数据集包含 6 万张手写数字的灰度图像。为了计算原因，我们随机抽取了 6000 张图像。数字图像有 $28 \times 28 = 784$ 个像素（即尺寸）。奥利维蒂面部数据集包含 40 位个体的图像，视角变化较小，表情变化大，偶尔会加戴眼镜。数据集包含 400 张图像（每个人 10 张），大小为 $92 \times 112 = 10,304$ 像素，并根据身份进行标记。COIL-20 数据集 (Nene 等, 1996) 包含 20 张图像

6. MNIST 数据集可从 <http://yann.lecun.com/exdb/mnist/index.html> 公开获取。

7. 奥利维蒂面孔数据集可从 <http://mambo.ucsc.edu/psl/olivetti.html> 公开获取。

different objects viewed from 72 equally spaced orientations, yielding a total of 1,440 images. The images contain $32 \times 32 = 1,024$ pixels.

4.2 Experimental Setup

In all of our experiments, we start by using PCA to reduce the dimensionality of the data to 30. This speeds up the computation of pairwise distances between the datapoints and suppresses some noise without severely distorting the interpoint distances. We then use each of the dimensionality reduction techniques to convert the 30-dimensional representation to a two-dimensional map and we show the resulting map as a scatterplot. For all of the data sets, there is information about the class of each datapoint, but the class information is only used to select a color and/or symbol for the map points. The class information is not used to determine the spatial coordinates of the map points. The coloring thus provides a way of evaluating how well the map preserves the similarities within each class.

The cost function parameter settings we employed in our experiments are listed in Table 1. In the table, *Perp* represents the perplexity of the conditional probability distribution induced by a Gaussian kernel and k represents the number of nearest neighbors employed in a neighborhood graph. In the experiments with Isomap and LLE, we only visualize datapoints that correspond to vertices in the largest connected component of the neighborhood graph.⁸ For the Sammon mapping optimization, we performed Newton’s method for 500 iterations.

<i>Technique</i>	<i>Cost function parameters</i>
t-SNE	<i>Perp</i> = 40
Sammon mapping	none
Isomap	$k = 12$
LLE	$k = 12$

Table 1: Cost function parameter settings for the experiments.

4.3 Results

In Figures 2 and 3, we show the results of our experiments with t-SNE, Sammon mapping, Isomap, and LLE on the MNIST data set. The results reveal the strong performance of t-SNE compared to the other techniques. In particular, Sammon mapping constructs a “ball” in which only three classes (representing the digits 0, 1, and 7) are somewhat separated from the other classes. Isomap and LLE produce solutions in which there are large overlaps between the digit classes. In contrast, t-SNE constructs a map in which the separation between the digit classes is almost perfect. Moreover, detailed inspection of the t-SNE map reveals that much of the local structure of the data (such as the orientation of the ones) is captured as well. This is illustrated in more detail in Section 5 (see Figure 7). The map produced by t-SNE contains some points that are clustered with the wrong class, but most of these points correspond to distorted digits many of which are difficult to identify.

Figure 4 shows the results of applying t-SNE, Sammon mapping, Isomap, and LLE to the Olivetti faces data set. Again, Isomap and LLE produce solutions that provide little insight into the class

8. Isomap and LLE require data that gives rise to a neighborhood graph that is connected.

从 72 个均匀分布的方向观察不同天体，总计 1440 张图像。图像包含 $32 \times 32=1,024$ 像素。

4.2 实验装置

在所有实验中，我们首先使用 PCA 将数据的维度降低到 30。这加快了数据点间两两距离的计算，并抑制了一些噪声，同时又不会严重扭曲点间距离。然后我们利用各种降维技术将 30 维表示转换为二维映射，并将生成的映射展示为散点图。对于所有数据集，都有关于每个数据点类别的信息，但类别信息仅用于为映射点选择颜色和/或符号。类别信息不用于确定地图点的空间坐标。因此，着色提供了评估地图在各类内相似性保持程度的方法。

我们在实验中使用的成本函数参数设置列于表 1。在表格中，Perp 表示由高斯核诱导的条件概率分布的困惑度，k 表示邻域图中使用的最近邻数量。在 Isomap 和 LLE 的实验中，我们只可视化对应邻域图最大连通分量顶点的数据点。对于萨蒙映射优化，我们进行了牛顿方法的 500 次迭代。

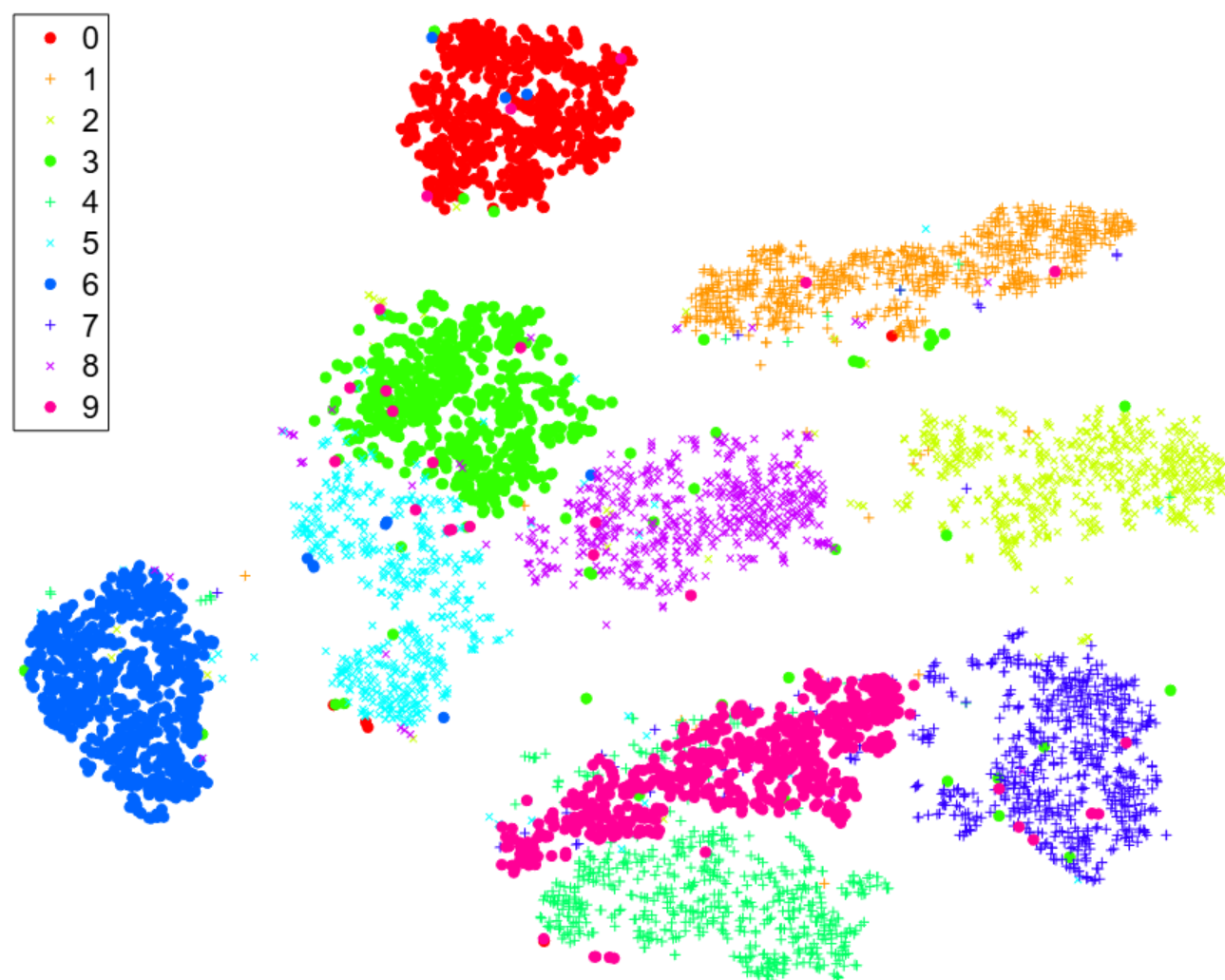
技术成本函数参数 t-SNE Perp = 40 Sammon 映射 无 等位	
映射 k = 12 LLE k = 12	

表 1: 实验成本函数参数设置。

4.3 结果

图 2 和图 3 展示了我们在 MNIST 数据集上使用 t-SNE、Sammon 映射、等位图和 LLE 实验的结果。结果显示 t-SNE 相较其他技术表现出优异的性能。特别地，萨蒙映射构造了一个“球体”，其中只有三个类别（代表数字 0、1 和 7）与其他类别有一定的分离。等值映射和 LLE 生成的解中，数字类别之间存在大量重叠。相比之下，tSNE 构建了一个位类间距几乎完美的映射。此外，对 t-SNE 地图的详细检查显示，数据的局部结构（如 1 的方向）也被捕捉到了。这一点在第 5 节（见图 7）中有更详细的说明。t-SNE 生成的映射包含一些与错误类别聚集的点，但大多数点对应的是扭曲数字，其中许多难以识别。图 4 展示了将 t-SNE、Sammon 映射、Isomap 和 LLE 应用于 Olivetti 人脸数据集的结果。同样，Isomap 和 LLE 的解答对该类的了解有限

8. 等位映射和 LLE 需要产生连通邻域图的数据。

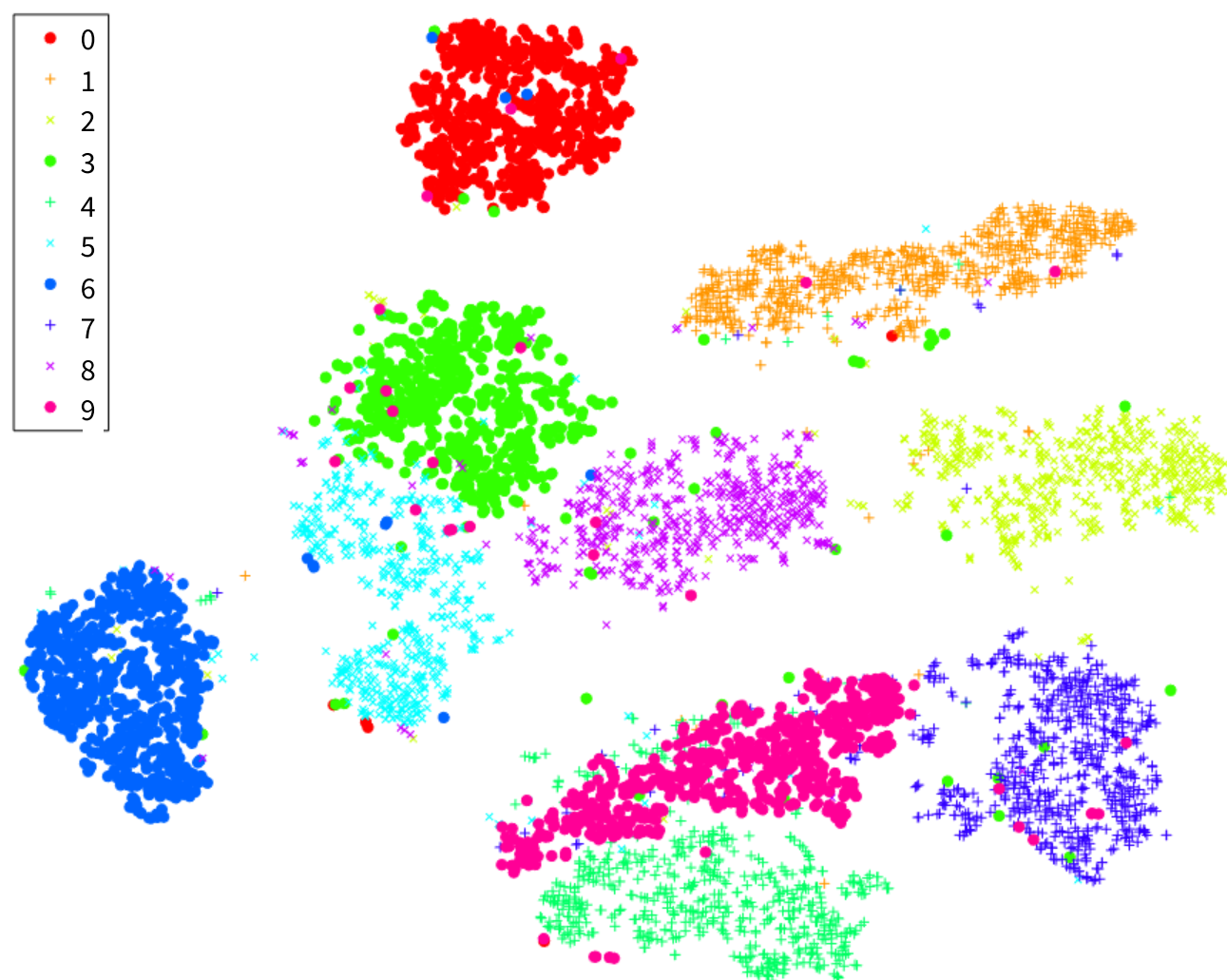


(a) Visualization by t-SNE.



(b) Visualization by Sammon mapping.

Figure 2: Visualizations of 6,000 handwritten digits from the MNIST data set.

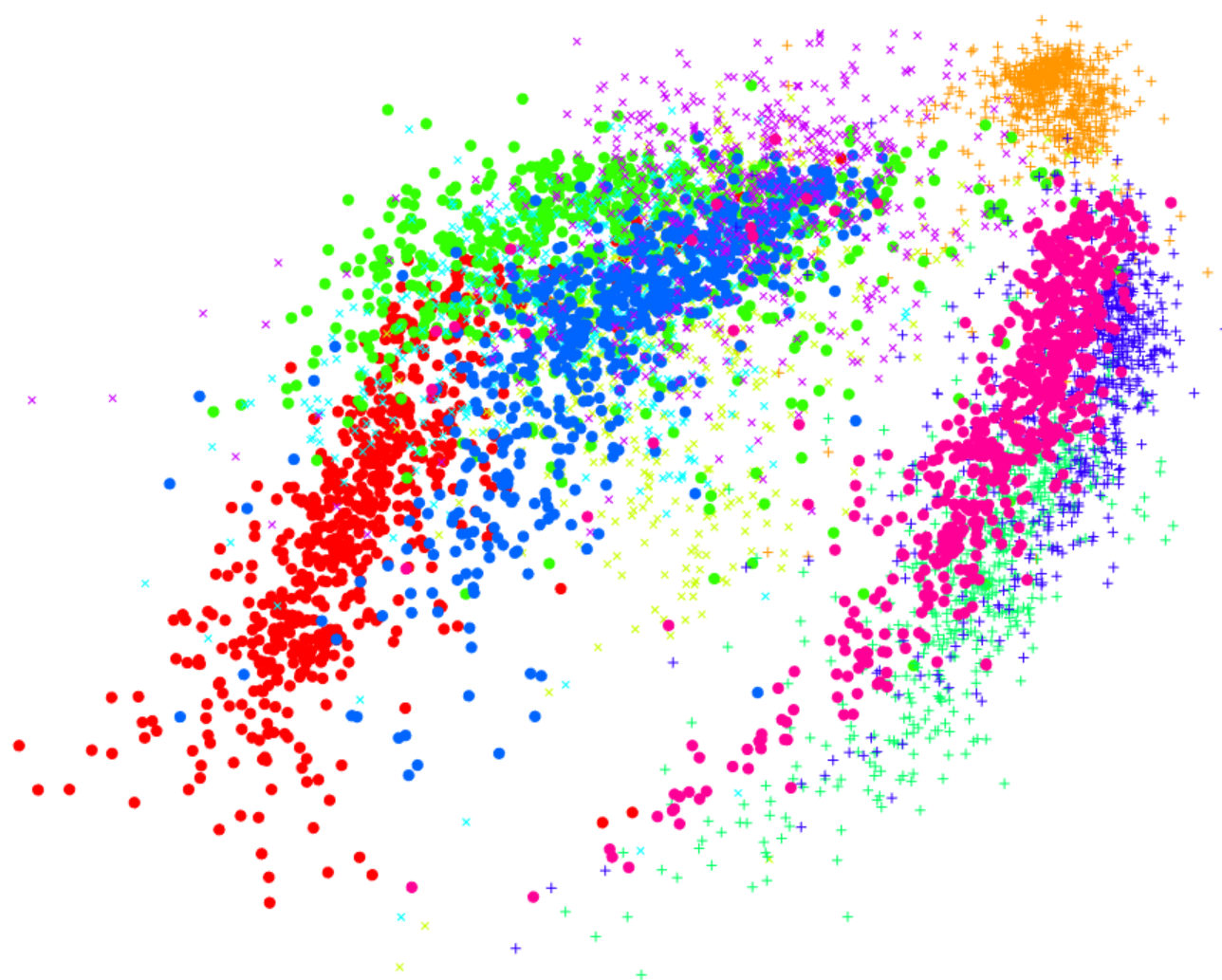


(a) 通过 t-SNE 进行可视化。

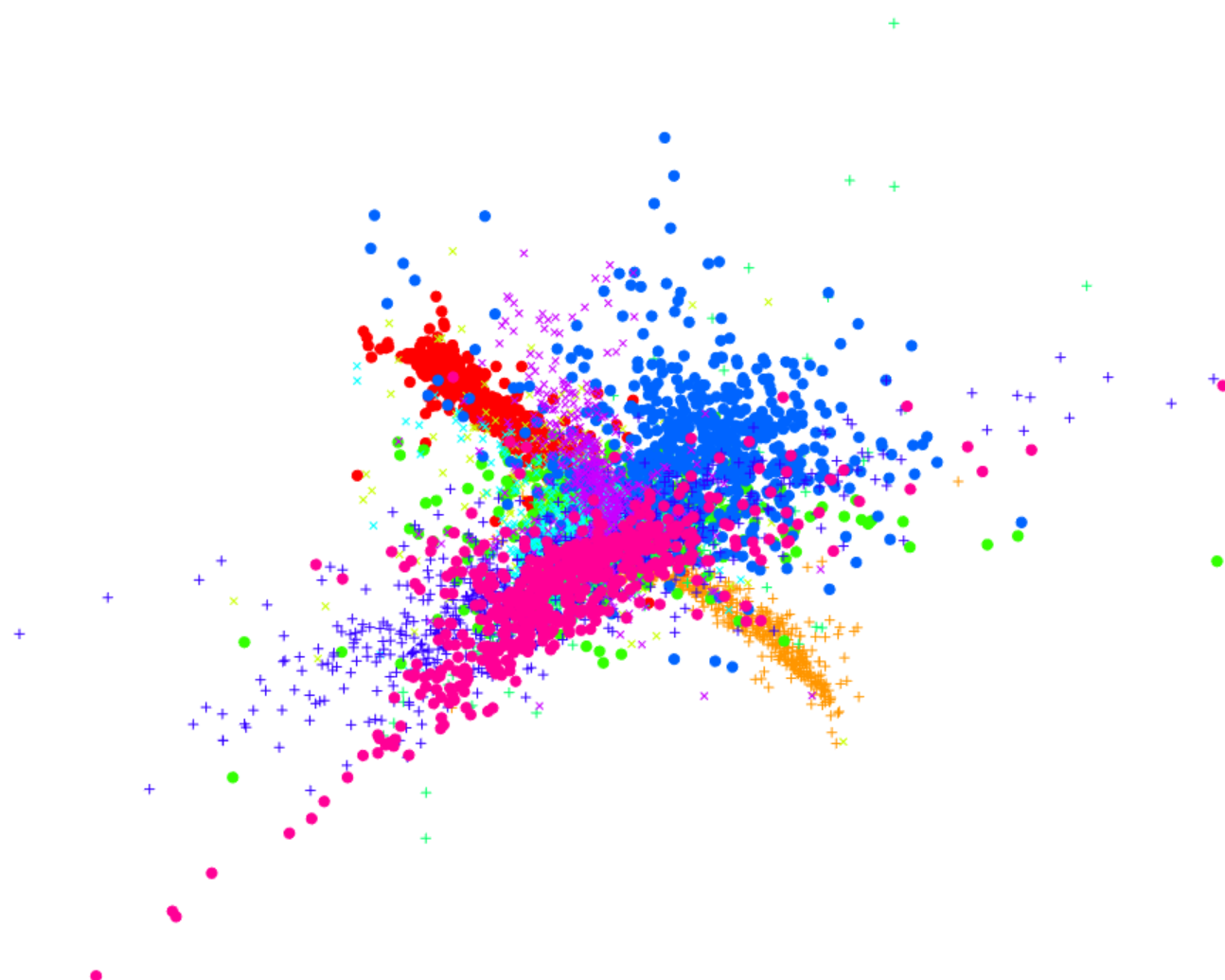


(b) 通过 Sammon 映射实现可视化。

图 2: 来自 MNIST 数据集的 6000 个手写数字的可视化。

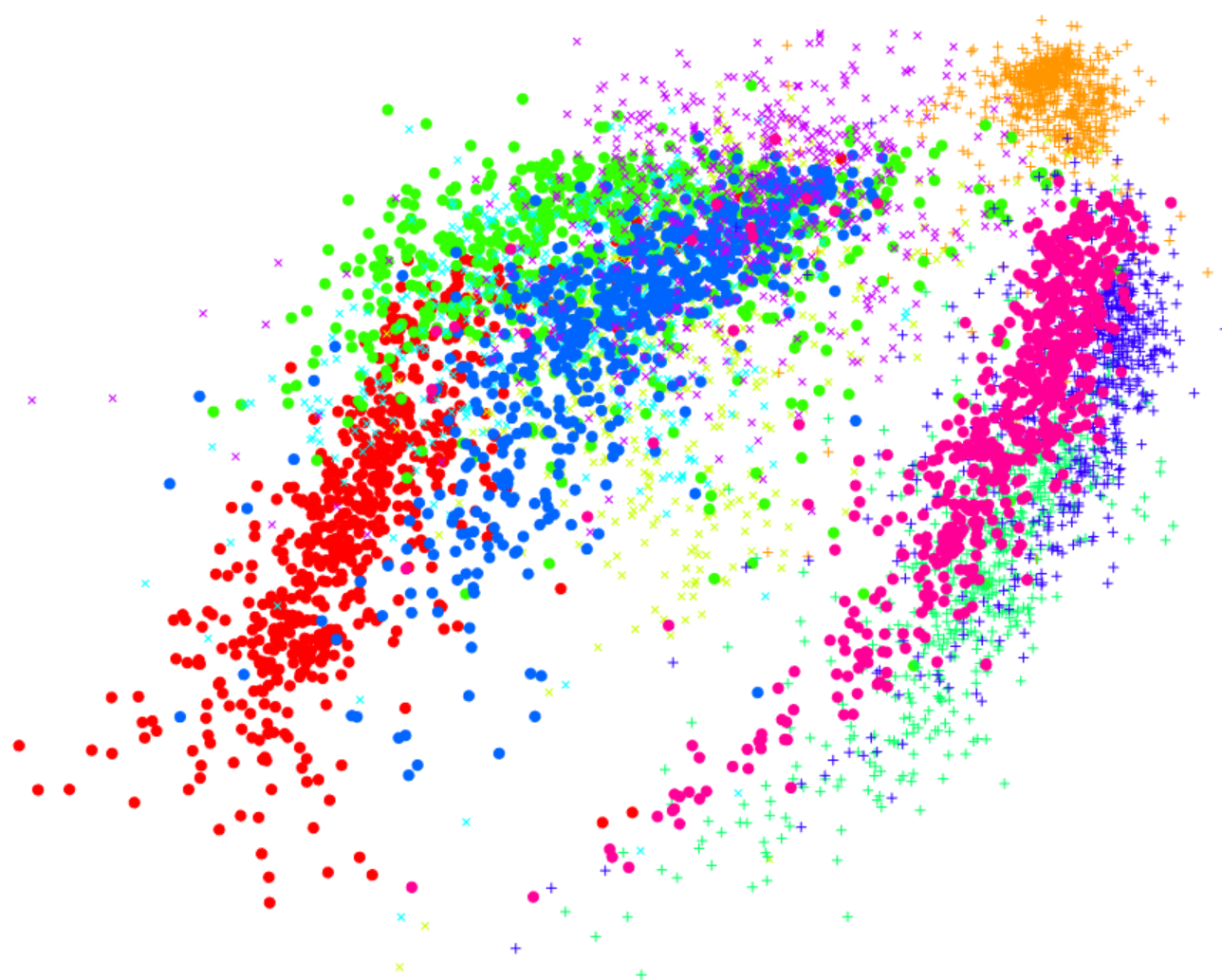


(a) Visualization by Isomap.

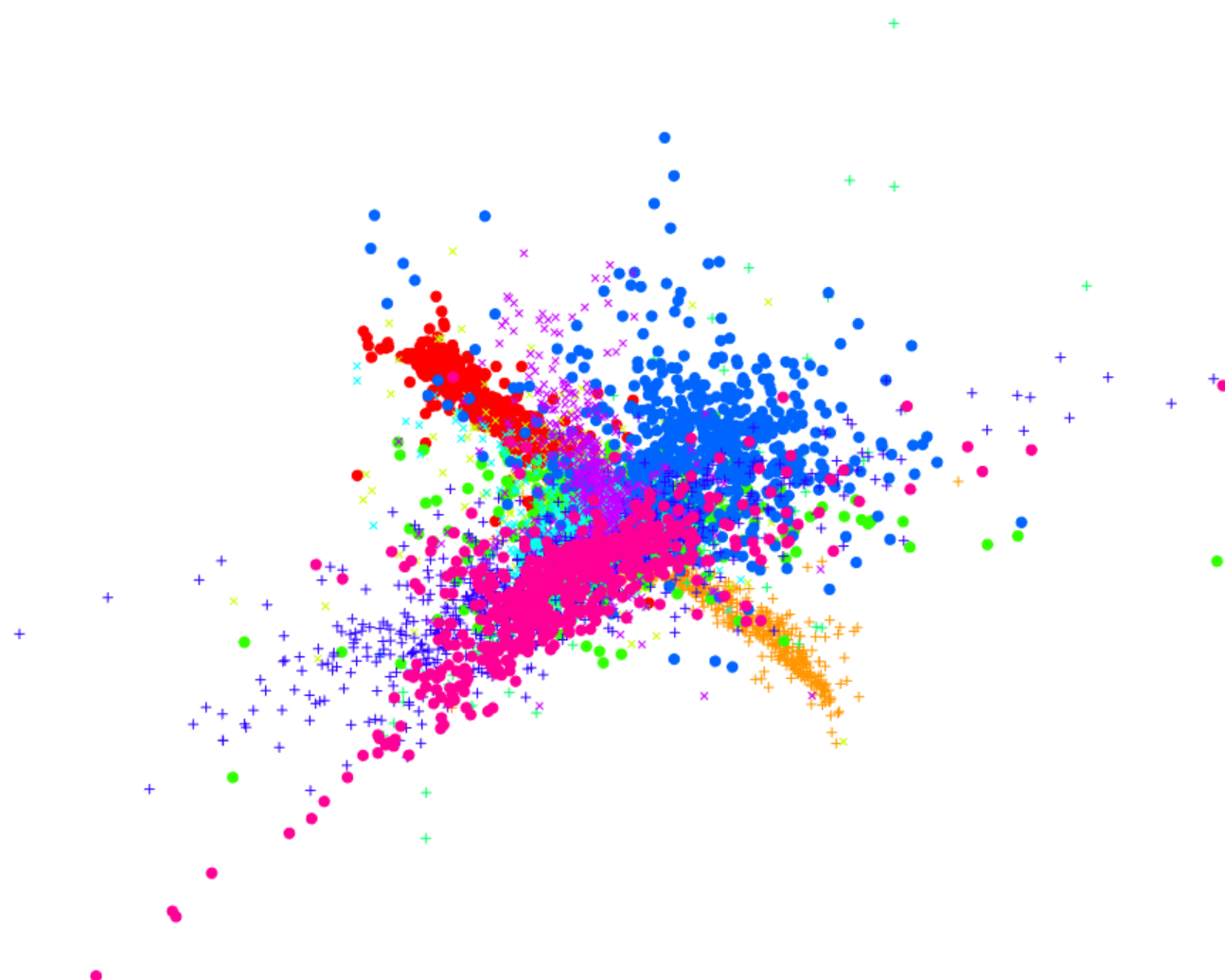


(b) Visualization by LLE.

Figure 3: Visualizations of 6,000 handwritten digits from the MNIST data set.



(a) 通过等值图进行可视化。



(b) LLE 的可视化。

图 3: 来自 MNIST 数据集的 6,000 个手写数字的可视化。

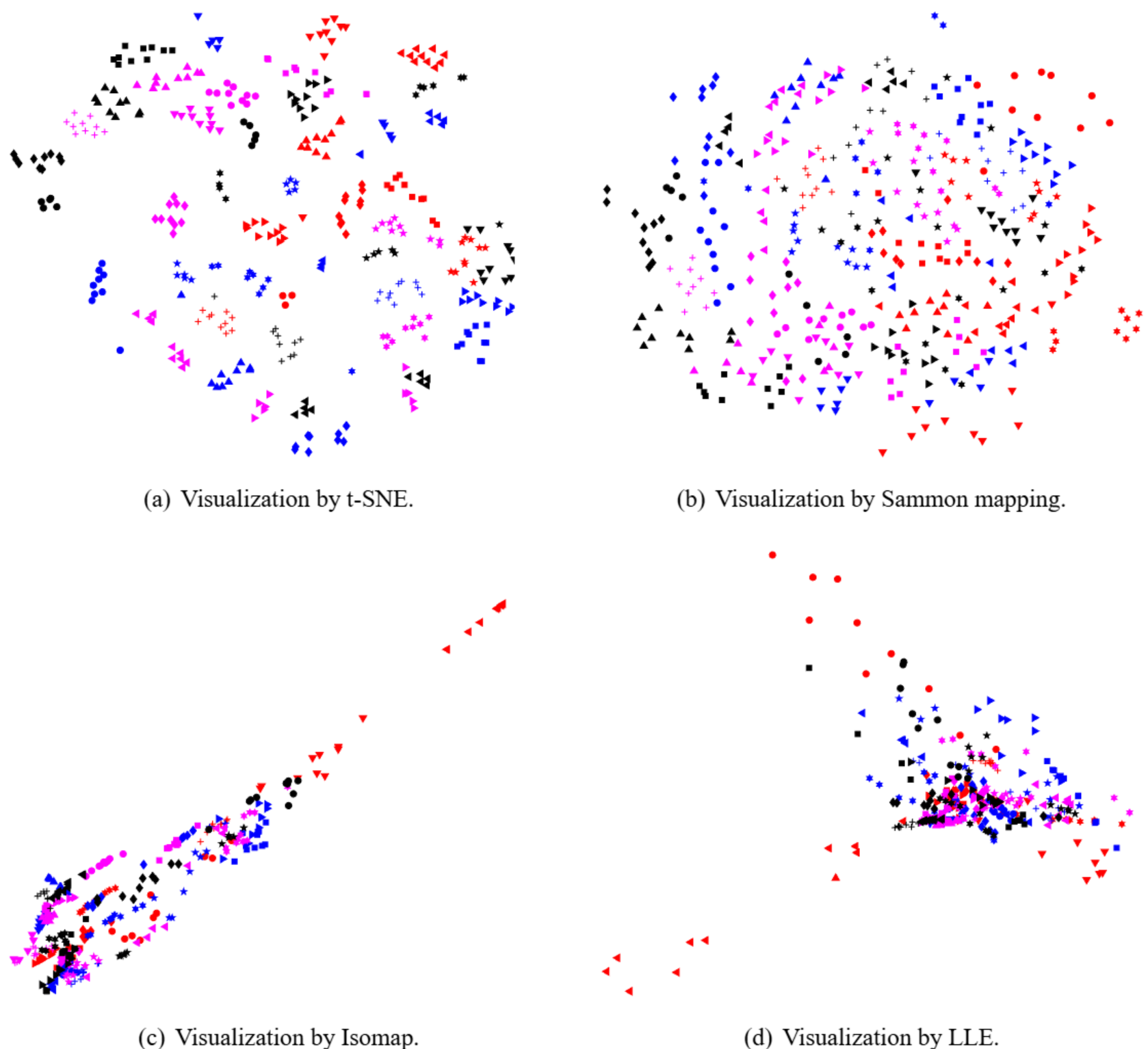
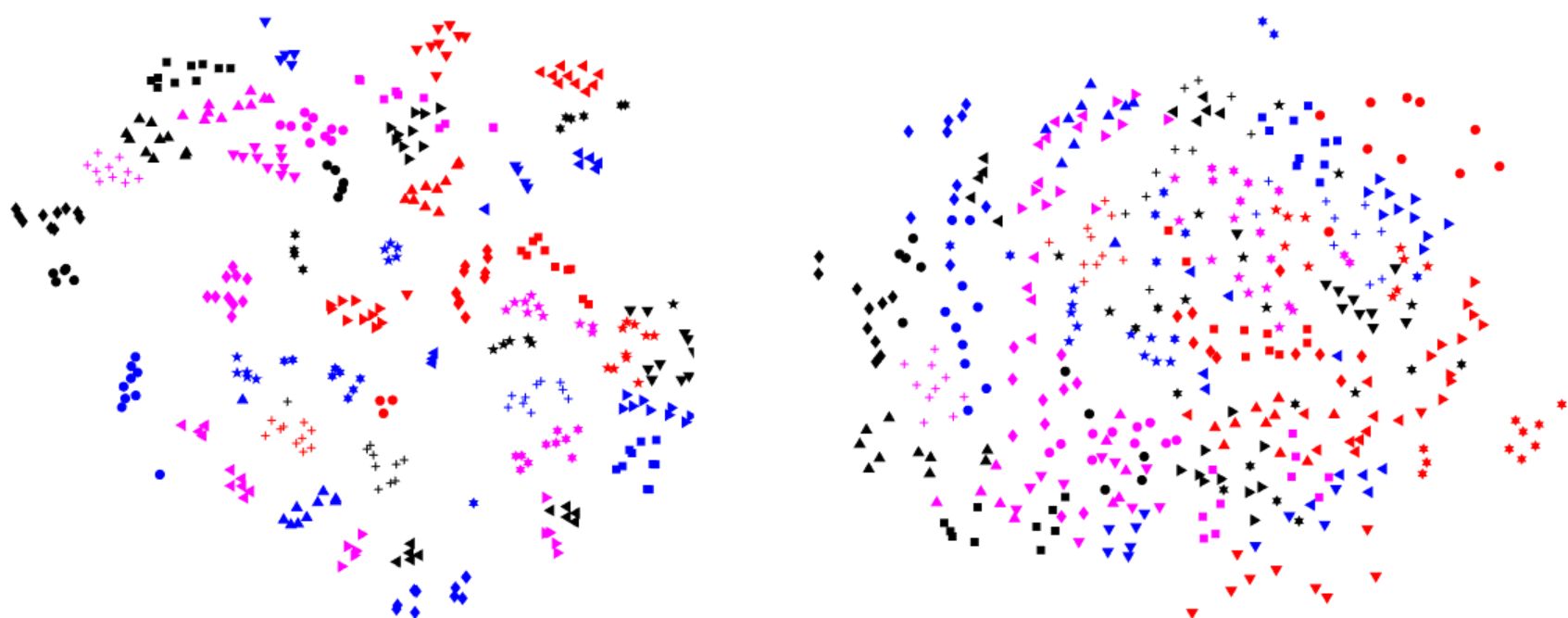


Figure 4: Visualizations of the Olivetti faces data set.

structure of the data. The map constructed by Sammon mapping is significantly better, since it models many of the members of each class fairly close together, but none of the classes are clearly separated in the Sammon map. In contrast, t-SNE does a much better job of revealing the natural classes in the data. Some individuals have their ten images split into two clusters, usually because a subset of the images have the head facing in a significantly different direction, or because they have a very different expression or glasses. For these individuals, it is not clear that their ten images form a natural class when using Euclidean distance in pixel space.

Figure 5 shows the results of applying t-SNE, Sammon mapping, Isomap, and LLE to the COIL-20 data set. For many of the 20 objects, t-SNE accurately represents the one-dimensional manifold of viewpoints as a closed loop. For objects which look similar from the front and the back, t-SNE distorts the loop so that the images of front and back are mapped to nearby points. For the four types of toy car in the COIL-20 data set (the four aligned “sausages” in the bottom-left of the t-SNE map), the four rotation manifolds are aligned by the orientation of the cars to capture the high



(a) 通过 t-SNE 进行可视化。(b) 通过 Sammon 映射实现可视化。



(c) 通过等距图进行可视化。(d) LLE 的可视化。

图 4：奥利维蒂人脸数据集的可视化。

数据结构。Sammon 地图构建的地图明显更好，因为它模拟了每个职业的许多成员，但 Sammon 地图上没有明显区分。相比之下，t-SNE 在揭示数据中的自然类方面做得更好。有些人的十张图像被分成两组，通常是因为部分图像的头部朝向显著不同，或者他们的表情或眼镜截然不同。对于这些个体来说，在使用像素空间中的欧几里得距离时，他们的十张图像是否构成一个自然类尚不明确。

图 5 展示了将 t-SNE、Sammon 映射、等位图和 LLE 应用于 COIL20 数据集的结果。对于 20 个对象中的许多，t-SNE 准确地表示一维视点流形为闭环。对于从正面和背面看起来相似的物体，t-SNE 会扭曲循环，使得正面和背面的图像映射到附近的点。对于 COIL-20 数据集中的四种玩具车（tSNE 图左下角的四个对齐“香肠”），四个旋转歧管根据车辆的方向对齐以捕捉高频

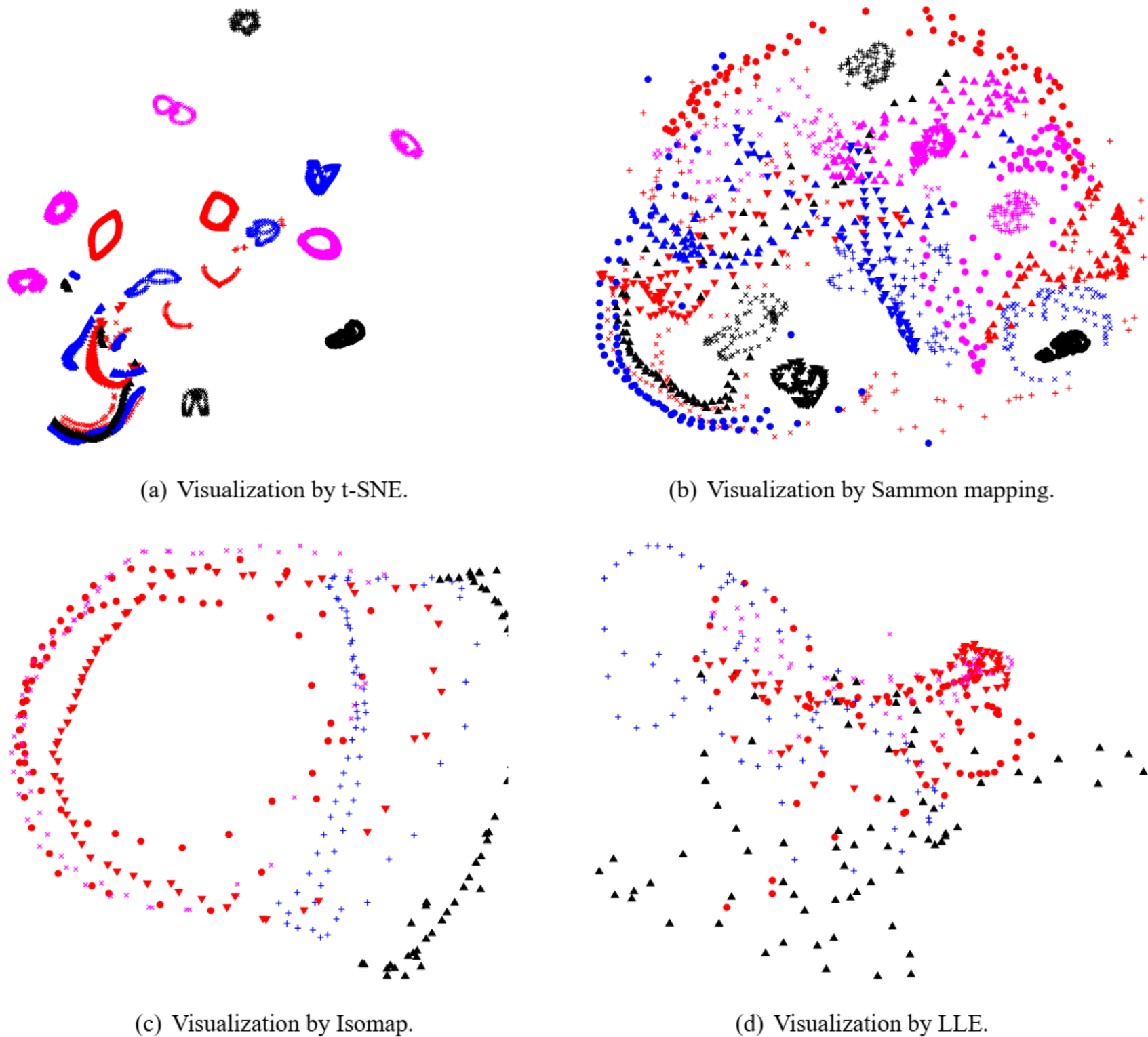
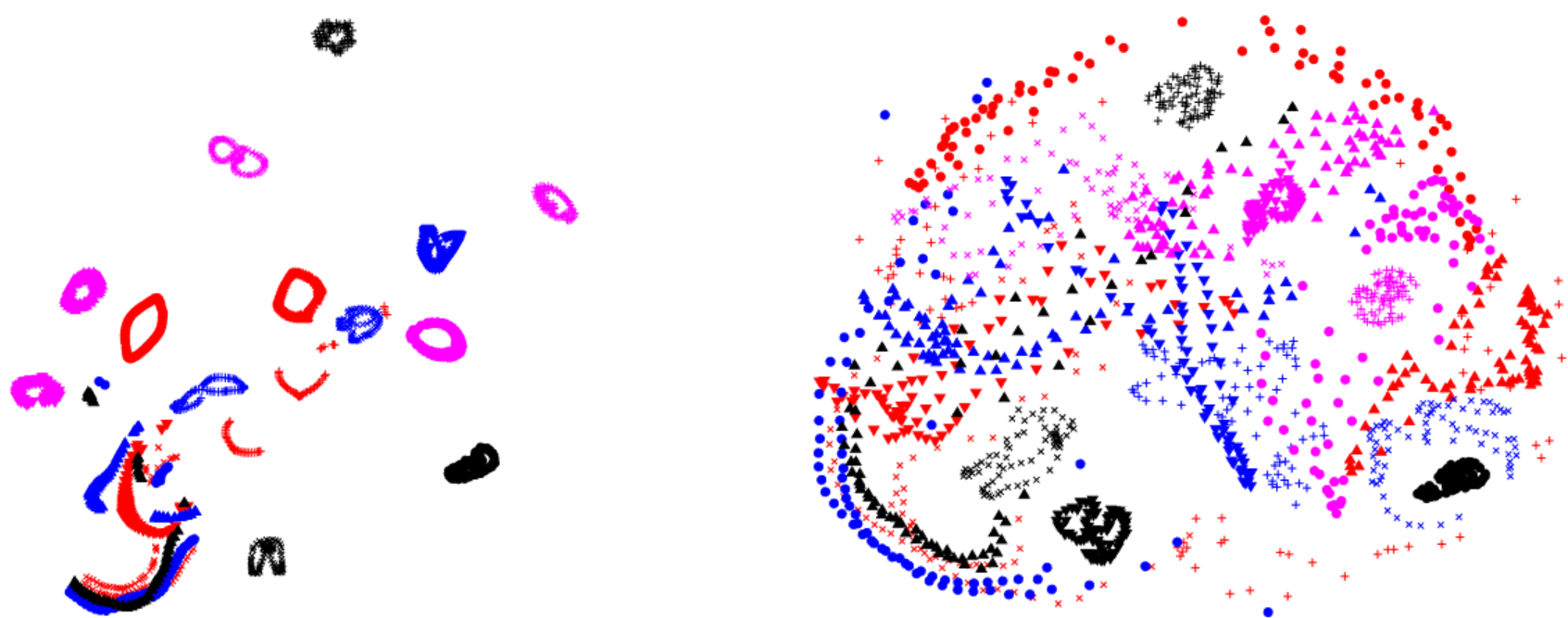


Figure 5: Visualizations of the COIL-20 data set.

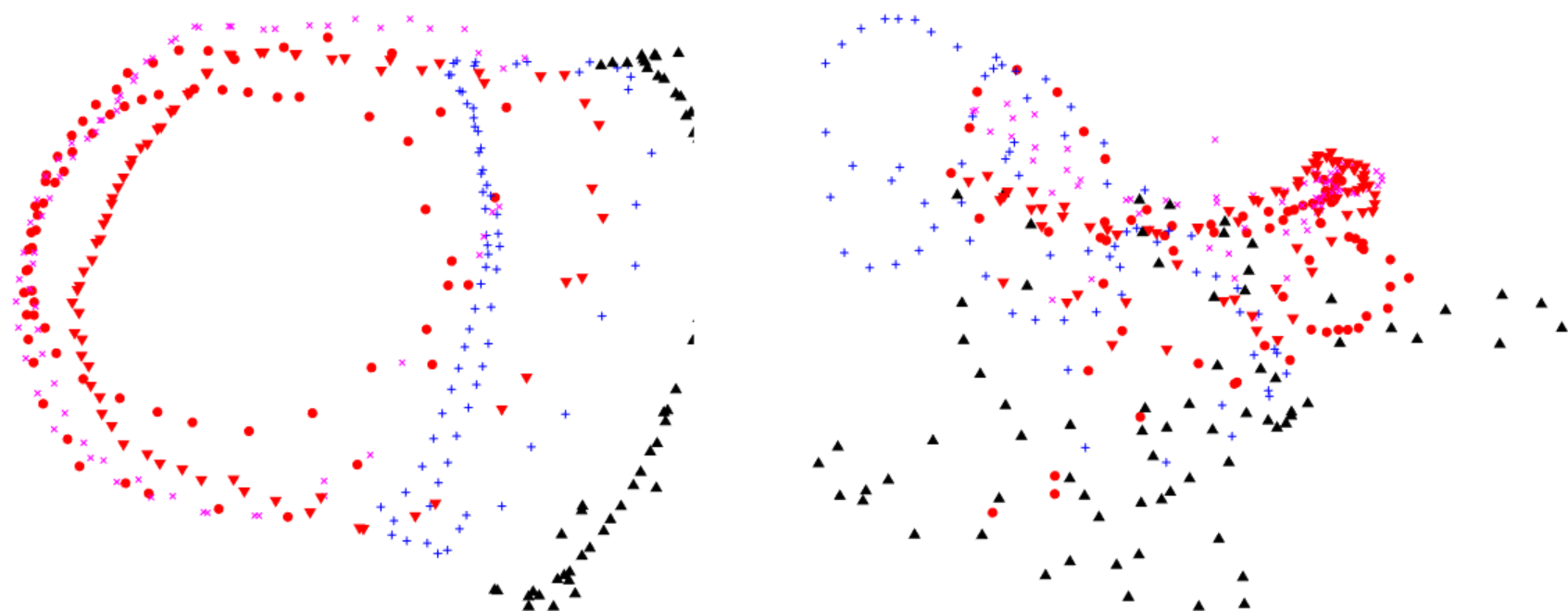
similarity between different cars at the same orientation. This prevents t-SNE from keeping the four manifolds clearly separate. Figure 5 also reveals that the other three techniques are not nearly as good at cleanly separating the manifolds that correspond to very different objects. In addition, Isomap and LLE only visualize a small number of classes from the COIL-20 data set, because the data set comprises a large number of widely separated submanifolds that give rise to small connected components in the neighborhood graph.

5. Applying t-SNE to Large Data Sets

Like many other visualization techniques, t-SNE has a computational and memory complexity that is quadratic in the number of datapoints. This makes it infeasible to apply the standard version of t-SNE to data sets that contain many more than, say, 10,000 points. Obviously, it is possible to pick a random subset of the datapoints and display them using t-SNE, but such an approach fails to



(a) 通过 t-SNE 进行可视化。(b) 通过 Sammon 映射实现可视化。



(c) 通过等距图进行可视化。(d) LLE 的可视化。

图 5: COIL-20 数据集的可视化。

以及 LLE 与 COIL20 数据集中不同车厢在相同方向上的相似度。这防止了 t-SNE 将四个流形清晰分开。图 5 还显示，其他三种技术在清晰分离对应截然不同对象的流形方面远不如前者。此外，Isomap 和 LLE 仅可视化 COIL-20 数据集中的少数类别，因为数据集包含大量分散的子流形，导致邻域图中产生小的连通分量。

5. 将 t-SNE 应用于大型数据集

像许多其他可视化技术一样，t-SNE 的计算和内存复杂度为数据点数的平方。这使得将标准版本的 t-SNE 应用于包含超过 10,000 点的数据集是不可行的。显然，可以随机选取数据点子集并用 t-SNE 显示，但这种方法无法实现

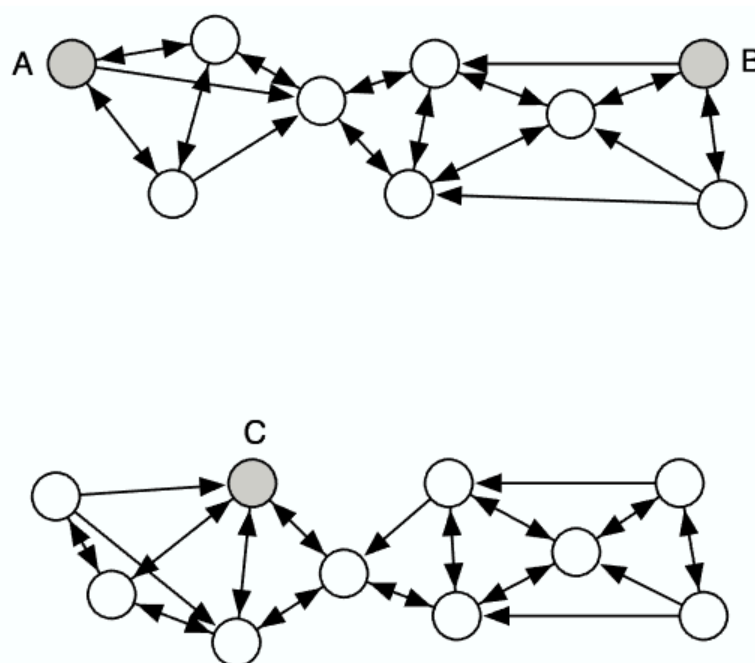
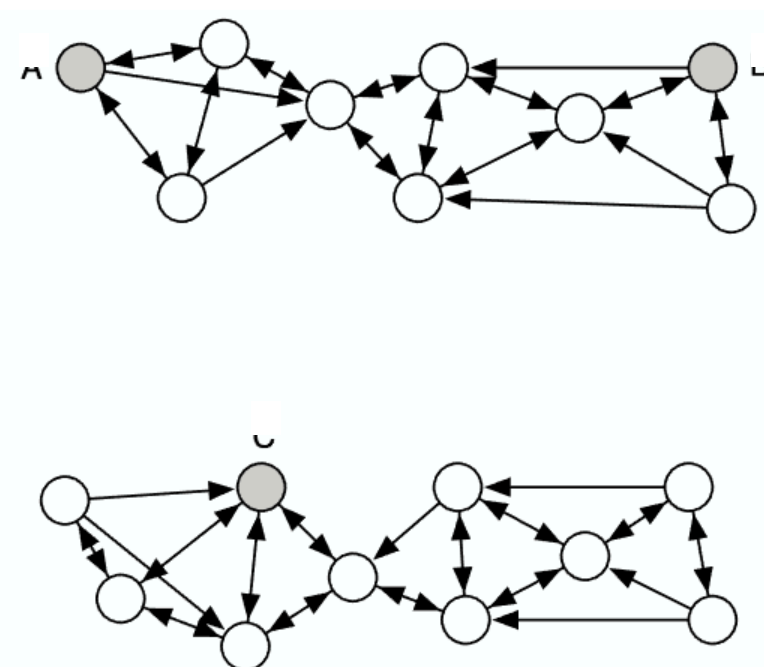


Figure 6: An illustration of the advantage of the random walk version of t-SNE over a standard landmark approach. The shaded points A, B, and C are three (almost) equidistant landmark points, whereas the non-shaded datapoints are non-landmark points. The arrows represent a directed neighborhood graph where $k = 3$. In a standard landmark approach, the pairwise affinity between A and B is approximately equal to the pairwise affinity between A and C. In the random walk version of t-SNE, the pairwise affinity between A and B is much larger than the pairwise affinity between A and C, and therefore, it reflects the structure of the data much better.

make use of the information that the undisplayed datapoints provide about the underlying manifolds. Suppose, for example, that A, B, and C are all equidistant in the high-dimensional space. If there are many undisplayed datapoints between A and B and none between A and C, it is much more likely that A and B are part of the same cluster than A and C. This is illustrated in Figure 6. In this section, we show how t-SNE can be modified to display a random subset of the datapoints (so-called landmark points) in a way that uses information from the entire (possibly very large) data set.

We start by choosing a desired number of neighbors and creating a neighborhood graph for all of the datapoints. Although this is computationally intensive, it is only done once. Then, for each of the landmark points, we define a random walk starting at that landmark point and terminating as soon as it lands on another landmark point. During a random walk, the probability of choosing an edge emanating from node x_i to node x_j is proportional to $e^{-\|x_i - x_j\|^2}$. We define $p_{j|i}$ to be the fraction of random walks starting at landmark point x_i that terminate at landmark point x_j . This has some resemblance to the way Isomap measures pairwise distances between points. However, as in diffusion maps (Lafon and Lee, 2006; Nadler et al., 2006), rather than looking for the shortest path through the neighborhood graph, the random walk-based affinity measure integrates over all paths through the neighborhood graph. As a result, the random walk-based affinity measure is much less sensitive to “short-circuits” (Lee and Verleysen, 2005), in which a single noisy datapoint provides a bridge between two regions of dataspace that should be far apart in the map. Similar approaches using random walks have also been successfully applied to, for example, semi-supervised learning (Szummer and Jaakkola, 2001; Zhu et al., 2003) and image segmentation (Grady, 2006).



t-SNE 的计算和内存复杂度为数据点数的平方。图 6：随机游走版本 t-SNE 相较标准模型优势的示例里程碑式的进度。阴影点 A、B 和 C 是三个（几乎）等距的地标点，而非阴影数据点则是非地标点。箭头表示一个有向邻域图，其中 $k = 3$ 。在标准的里程碑方法中，A 和 B 之间的成对亲和力大致等于 A 和 C 的成对亲和力。在随机游走的 t-SNE 版本中，A 和 B 之间的两对亲和力远大于 A 和 C 之间的亲和力，因此更能反映数据结构。

利用未显示数据点提供的关于底层流形的信息。例如，假设 A、B 和 C 在高维空间中均等距。如果 A 和 B 之间有许多未显示的数据点，而 A 和 C 之间没有，那么 A 和 B 比 A 和 C 更可能属于同一簇。图 6 展示了这一点。本节将展示如何修改 t-SNE，以利用整个（可能非常庞大）数据集的信息来显示随机子集的数据点（所谓的里程碑点）。

我们首先选择一个目标数量的邻居，并为所有数据点创建一个邻域图。虽然计算量大，但只需完成一次。然后，对于每个地标点，我们定义一个随机游走，从该地标点开始，停在另一个地标点后立即终止。在随机游走过程中，选择从节点 x 到节点 y 的边的概率与 e 成正比。我们将 p 定义为从里程碑点 x 开始、终止于里程碑点 y 的随机漫步比例。这与 Isomap 测量点间成对距离的方式有些相似。然而，如扩散图 (Lafon 和 Lee, 2006; Nadler 等, 2006)，随机游走亲和度不是寻找邻域图中最短路径，而是对邻域图中所有路径进行积分。因此，基于随机游走的亲和度对“短路”的敏感度较低 (Lee 和 Verleysen, 2005)，即单个噪声数据点为地图上两个本应相距较远的数据空间区域提供桥梁。类似的随机游走方法已成功应用于半监督学习 (Szummer 和 Jaakkola, 2001; Zhu 等, 2003) 以及图像分割 (Grady, 2006)。

The most obvious way to compute the random walk-based similarities $p_{j|i}$ is to explicitly perform the random walks on the neighborhood graph, which works very well in practice, given that one can easily perform one million random walks per second. Alternatively, Grady (2006) presents an analytical solution to compute the pairwise similarities $p_{j|i}$ that involves solving a sparse linear system. The analytical solution to compute the similarities $p_{j|i}$ is sketched in Appendix B. In preliminary experiments, we did not find significant differences between performing the random walks explicitly and the analytical solution. In the experiment we present below, we explicitly performed the random walks because this is computationally less expensive. However, for very large data sets in which the landmark points are very sparse, the analytical solution may be more appropriate.

Figure 7 shows the results of an experiment, in which we applied the random walk version of t-SNE to 6,000 randomly selected digits from the MNIST data set, using all 60,000 digits to compute the pairwise affinities $p_{j|i}$. In the experiment, we used a neighborhood graph that was constructed using a value of $k = 20$ nearest neighbors.⁹ The inset of the figure shows the same visualization as a scatterplot in which the colors represent the labels of the digits. In the t-SNE map, all classes are clearly separated and the “continental” sevens form a small separate cluster. Moreover, t-SNE reveals the main dimensions of variation within each class, such as the orientation of the ones, fours, sevens, and nines, or the “loopiness” of the twos. The strong performance of t-SNE is also reflected in the generalization error of nearest neighbor classifiers that are trained on the low-dimensional representation. Whereas the generalization error (measured using 10-fold cross validation) of a 1-nearest neighbor classifier trained on the original 784-dimensional datapoints is 5.75%, the generalization error of a 1-nearest neighbor classifier trained on the two-dimensional data representation produced by t-SNE is only 5.13%. The computational requirements of random walk t-SNE are reasonable: it took only one hour of CPU time to construct the map in Figure 7.

6. Discussion

The results in the previous two sections (and those in the supplemental material) demonstrate the performance of t-SNE on a wide variety of data sets. In this section, we discuss the differences between t-SNE and other non-parametric techniques (Section 6.1), and we also discuss a number of weaknesses and possible improvements of t-SNE (Section 6.2).

6.1 Comparison with Related Techniques

Classical scaling (Torgerson, 1952), which is closely related to PCA (Mardia et al., 1979; Williams, 2002), finds a linear transformation of the data that minimizes the sum of the squared errors between high-dimensional pairwise distances and their low-dimensional representatives. A linear method such as classical scaling is not good at modeling curved manifolds and it focuses on preserving the distances between widely separated datapoints rather than on preserving the distances between nearby datapoints. An important approach that attempts to address the problems of classical scaling is the Sammon mapping (Sammon, 1969) which alters the cost function of classical scaling by dividing the squared error in the representation of each pairwise Euclidean distance by the original Euclidean distance in the high-dimensional space. The resulting cost function is given by

$$C = \frac{1}{\sum_{i,j} \|x_i - x_j\|} \sum_{i \neq j} \frac{(\|x_i - x_j\| - \|y_i - y_j\|)^2}{\|x_i - x_j\|},$$

9. In preliminary experiments, we found the performance of random walk t-SNE to be very robust under changes of k .

计算基于随机游走相似性 p 的最明显方法是显式地对邻域图进行随机游走，这在实际中非常有效，因为每秒可以轻松执行一百万次随机游走。另外，Grady (2006) 提出了一种解析解法，用于计算两对相似性 p ，涉及求解稀疏线性系统。计算相似性 p 的解析解见附录 B。在初步实验中，我们未发现显式随机游走与解析解之间存在显著差异。在下面展示的实验中，我们显式地执行了随机漫步，因为这样计算成本更低。然而，对于非常大的数据集，且标志点非常稀疏，分析解可能更为合适。

图 7 展示了一项实验的结果，我们将 t-SNE 的随机游走版本应用于 MNIST 数据集中随机选出的 60,000 个数字，计算出成对亲和力 p 。实验中，我们使用了一个邻域图，取值为 $k = 20$ 个最近邻。图的内嵌展示了与散点图相同的可视化，其中颜色代表数字标签。在 t-SNE 地图中，所有类别都被明确区分开来，“大陆”七组成了一个小而独立的集群。此外，t-SNE 揭示了每个类别内主要的变异维度，如一、四、七和九的方向，或二的“环状度”。t-SNE 的强劲性能也反映在训练于低维表示的最近邻分类器的泛化误差上。而在原始 784 维数据点上训练的 1 最近邻分类器，其泛化误差（通过 10 折交叉验证测量）为 5.75%；而用 t-SNE 生成的二维数据表示训练的 1 最近邻分类器的泛化误差仅为 5.13%。随机游走 t-SNE 的计算需求合理：构建图 7 中的映射仅用了一小时的 CPU 时间。

6. 讨论

前两节（及补充材料中）的结果展示了 t-SNE 在多种数据集上的表现。本节讨论 t-SNE 与其他非参数技术的区别（第 6.1 节），并讨论 t-SNE 的若干弱点及可能改进之处（第 6.2 节）。

6.1 与相关技术的比较

经典尺度 (Torgerson, 1952)，与 PCA 密切相关 (Mardia 等, 1979; Williams, 2002) 找到了一种线性变换，使高维成对距离与其低维代表值之间的平方误差之和最小化。线性方法如经典缩放法不适合建模曲率流形，它更注重保持距离较远的数据点之间的距离，而非保持相邻数据点之间的距离。一种试图解决经典缩放问题的重要方法是 Sammon 映射 (Sammon, 1969)，它通过将每个两对欧几里得距离表示的平方误差除以高维空间中的原始欧几里得距离，从而改变经典缩放的代价函数。所得成本函数为

$$C = \frac{1}{\sum \|x - x\|} \sum_{i,j} \frac{(\|x - x\| - \|y - y\|)^2}{\|x - x\|},$$

9. 在初步实验中，我们发现随机游走 t-SNE 在 k 变化下表现非常稳健。

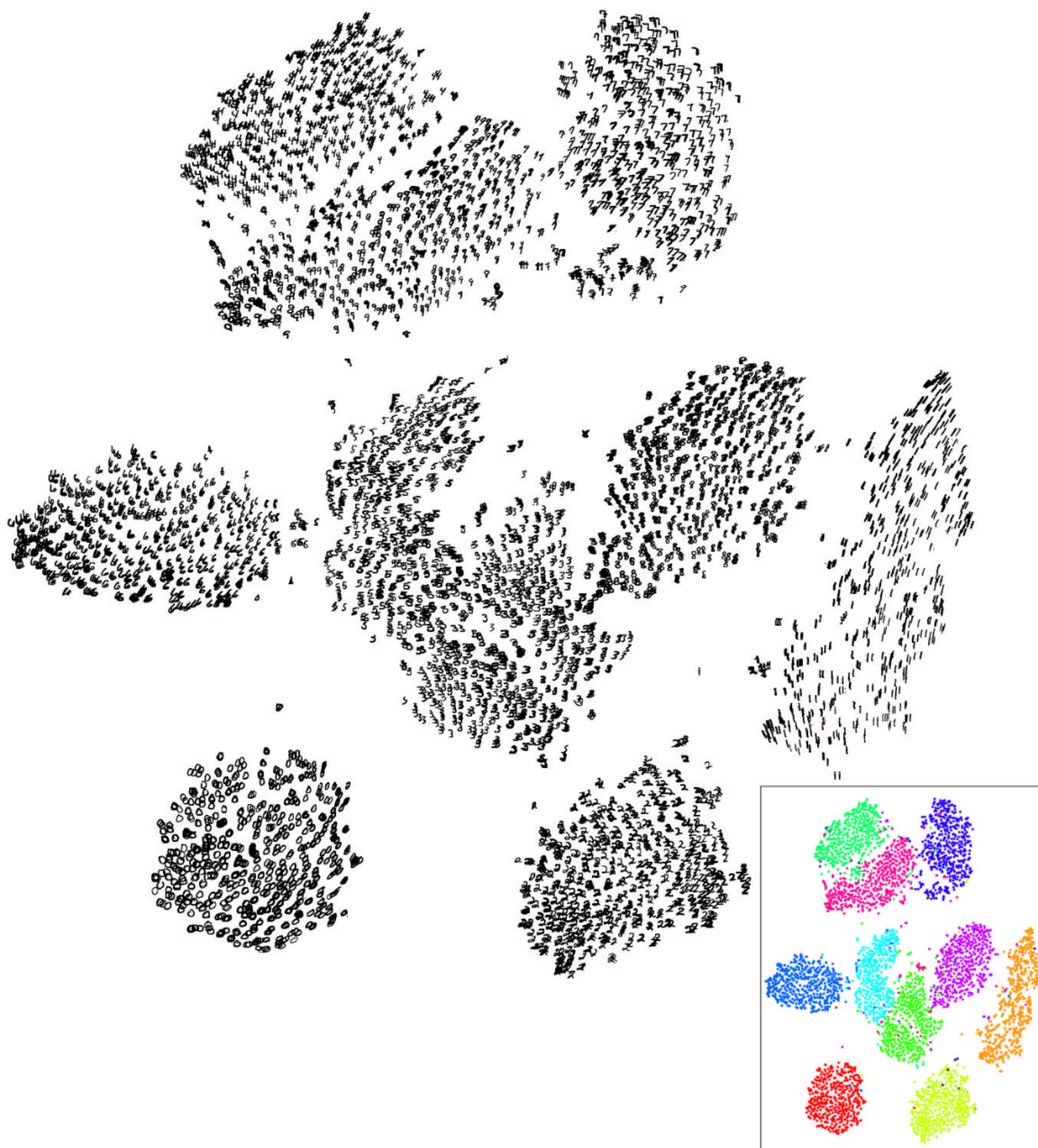


Figure 7: Visualization of 6,000 digits from the MNIST data set produced by the random walk version of t-SNE (employing all 60,000 digit images).

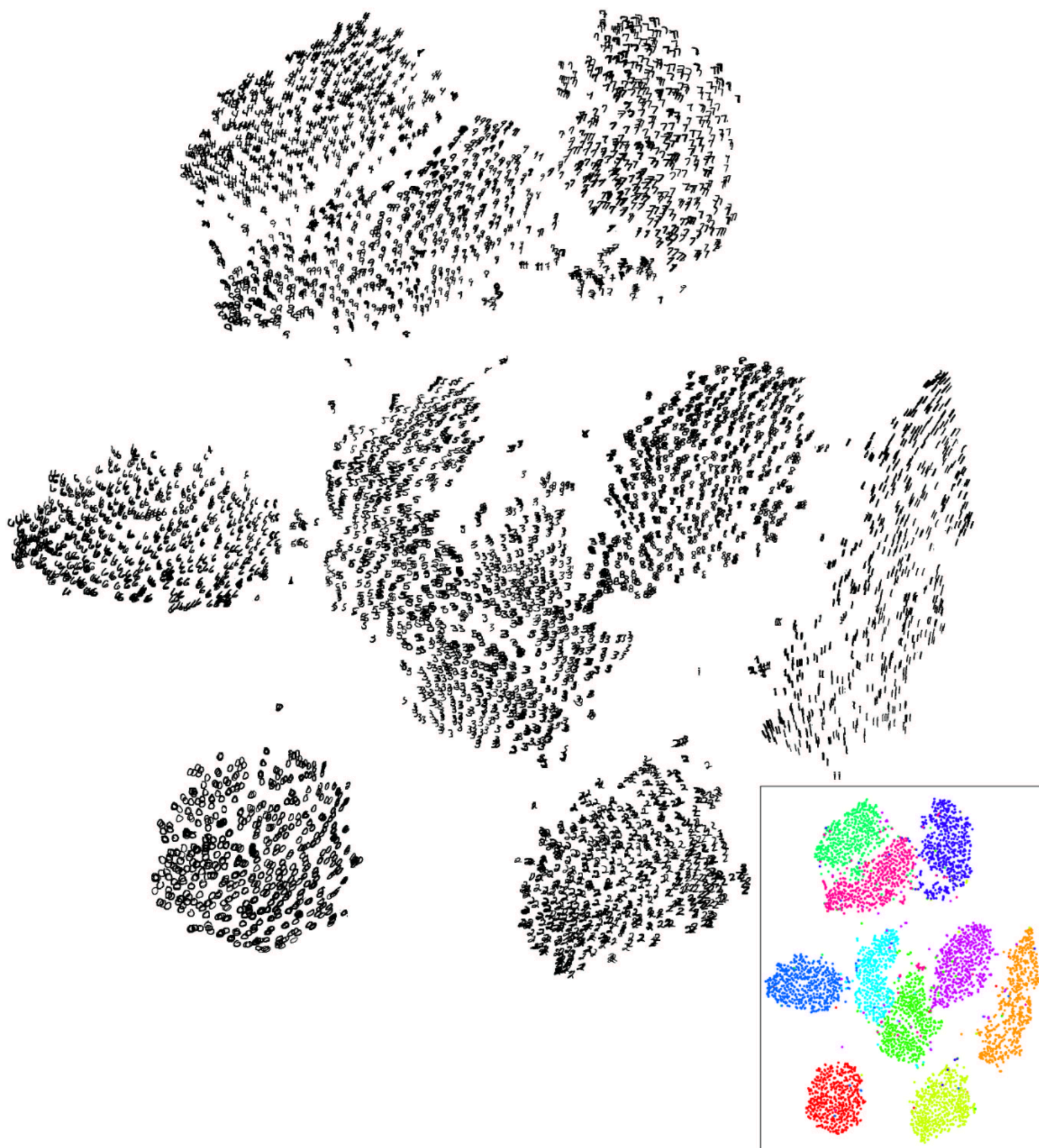


图 7: 由随机游走版本 t-SNE 生成的 MNIST 数据集中 6000 位数字的可视化 (使用全部 60000 位图像)。

where the constant outside of the sum is added in order to simplify the derivation of the gradient. The main weakness of the Sammon cost function is that the importance of retaining small pairwise distances in the map is largely dependent on small differences in these pairwise distances. In particular, a small error in the model of two high-dimensional points that are extremely close together results in a large contribution to the cost function. Since all small pairwise distances constitute the local structure of the data, it seems more appropriate to aim to assign approximately equal importance to all small pairwise distances.

In contrast to Sammon mapping, the Gaussian kernel employed in the high-dimensional space by t-SNE defines a soft border between the local and global structure of the data and for pairs of datapoints that are close together relative to the standard deviation of the Gaussian, the importance of modeling their separations is almost independent of the magnitudes of those separations. Moreover, t-SNE determines the local neighborhood size for each datapoint separately based on the local density of the data (by forcing each conditional probability distribution P_i to have the same perplexity).

The strong performance of t-SNE compared to Isomap is partly explained by Isomap's susceptibility to "short-circuiting". Also, Isomap mainly focuses on modeling large geodesic distances rather than small ones.

The strong performance of t-SNE compared to LLE is mainly due to a basic weakness of LLE: the only thing that prevents all datapoints from collapsing onto a single point is a constraint on the covariance of the low-dimensional representation. In practice, this constraint is often satisfied by placing most of the map points near the center of the map and using a few widely scattered points to create large covariance (see Figure 3(b) and 4(d)). For neighborhood graphs that are almost disconnected, the covariance constraint can also be satisfied by a "curdled" map in which there are a few widely separated, collapsed subsets corresponding to the almost disconnected components. Furthermore, neighborhood-graph based techniques (such as Isomap and LLE) are not capable of visualizing data that consists of two or more widely separated submanifolds, because such data does not give rise to a connected neighborhood graph. It is possible to produce a separate map for each connected component, but this loses information about the relative similarities of the separate components.

Like Isomap and LLE, the random walk version of t-SNE employs neighborhood graphs, but it does not suffer from short-circuiting problems because the pairwise similarities between the high-dimensional datapoints are computed by integrating over all paths through the neighborhood graph. Because of the diffusion-based interpretation of the conditional probabilities underlying the random walk version of t-SNE, it is useful to compare t-SNE to diffusion maps. Diffusion maps define a "diffusion distance" on the high-dimensional datapoints that is given by

$$D^{(t)}(x_i, x_j) = \sqrt{\sum_k \frac{(p_{ik}^{(t)} - p_{jk}^{(t)})^2}{\psi(x_k)^{(0)}}},$$

where $p_{ij}^{(t)}$ represents the probability of a particle traveling from x_i to x_j in t timesteps through a graph on the data with Gaussian emission probabilities. The term $\psi(x_k)^{(0)}$ is a measure for the local density of the points, and serves a similar purpose to the fixed perplexity Gaussian kernel that is employed in SNE. The diffusion map is formed by the principal non-trivial eigenvectors of the Markov matrix of the random walks of length t . It can be shown that when all $(n - 1)$ non-trivial eigenvec-

其中加了和外的常数，以简化梯度的推导。萨蒙成本函数的主要弱点在于，在地图中保留小的两两距离在很大程度上取决于这些两两距离的微小差异。特别地，两个极其接近的高维点模型中的小误差会导致成本函数产生较大贡献。由于所有小的成对距离构成了数据的局部结构，因此更合适的做法是将所有小成对距离赋予大致相同的重要性。

与萨蒙映射不同，t-SNE 在高维空间中使用的高斯核定义了数据局部与全局结构之间的软边界，对于相对于高斯标准差相近的数据点对，建模它们的分离的重要性几乎与这些分离的大小无关。此外，t-SNE 根据数据的局部密度分别确定每个数据点的局部邻域大小（通过强制每个条件概率分布 P_{t0} 具有相同的复杂度）。

t-SNE 优于同位图的优异性能部分归因于同位图易发生“短路”现象。此外，Isomap 主要专注于建模较大的测地线距离，而非较小的测地线距离。

t-SNE 相较 LLE 表现优异，主要源于 LLE 的一个基本弱点：唯一阻止所有数据点集中到单一点的，是对低维表示协方差的约束。实际上，通常通过将大部分映射点置于地图中心附近，并使用少数分散点来产生较大的协方差来满足这一约束（见图 3（b）和 4（d））。对于几乎不连通的邻域图，协方差约束也可以通过一个“凝结”映射满足，其中存在几个相距较远、坍塌的子集对应这些几乎不连通的分量。此外，基于邻域图的技术（如 Isomap 和 LLE）无法可视化由两个或更多相距较远的子流形组成的数据，因为此类数据不会产生连通邻域图。可以为每个连通分量制作独立映射，但这会丢失分量间相对相似性的信息。

与 Isomap 和 LLE 类似，t-SNE 的随机游走版本使用邻域图，但由于高维数据点之间的两两相似性是通过邻域图中所有路径积分计算的，因此不存在短路问题。由于基于扩散的条件概率解释随机游走版本 t-SNE 的条件概率，比较 t-SNE 与扩散映射是有用的。扩散映射定义了高维数据点上的“扩散距离”，由下式给出

$$D(x, x) = \sqrt{\frac{1}{\sum_{x'} \psi(x')}} \sqrt{\frac{p_{x,x}}{\psi(x)}},$$

其中 p 表示粒子从 x 到 x 在 t 个时间步内通过数据图，且其发射概率为高斯发射概率。项 $\psi(x)$ 是点的局部密度度量，功能类似于 SNE 中使用的固定迷惑高斯核。扩散映射由长度为 t 的随机游走的马尔可夫矩阵的主要非平凡特征向量构成。可以证明，当所有 $(n-1)$ 非平凡特征维克-

tors are employed, the Euclidean distances in the diffusion map are equal to the diffusion distances in the high-dimensional data representation (Lafon and Lee, 2006). Mathematically, diffusion maps minimize

$$C = \sum_i \sum_j \left(D^{(t)}(x_i, x_j) - \|y_i - y_j\| \right)^2.$$

As a result, diffusion maps are susceptible to the same problems as classical scaling: they assign much higher importance to modeling the large pairwise diffusion distances than the small ones and as a result, they are not good at retaining the local structure of the data. Moreover, in contrast to the random walk version of t-SNE, diffusion maps do not have a natural way of selecting the length, t , of the random walks.

In the supplemental material, we present results that reveal that t-SNE outperforms CCA (Demartines and Hérault, 1997), MVU (Weinberger et al., 2004), and Laplacian Eigenmaps (Belkin and Niyogi, 2002) as well. For CCA and the closely related CDA (Lee et al., 2000), these results can be partially explained by the hard border λ that these techniques define between local and global structure, as opposed to the soft border of t-SNE. Moreover, within the range λ , CCA suffers from the same weakness as Sammon mapping: it assigns extremely high importance to modeling the distance between two datapoints that are extremely close.

Like t-SNE, MVU (Weinberger et al., 2004) tries to model all of the small separations well but MVU insists on modeling them perfectly (i.e., it treats them as constraints) and a single erroneous constraint may severely affect the performance of MVU. This can occur when there is a short-circuit between two parts of a curved manifold that are far apart in the intrinsic manifold coordinates. Also, MVU makes no attempt to model longer range structure: It simply pulls the map points as far apart as possible subject to the hard constraints so, unlike t-SNE, it cannot be expected to produce sensible large-scale structure in the map.

For Laplacian Eigenmaps, the poor results relative to t-SNE may be explained by the fact that Laplacian Eigenmaps have the same covariance constraint as LLE, and it is easy to cheat on this constraint.

6.2 Weaknesses

Although we have shown that t-SNE compares favorably to other techniques for data visualization, t-SNE has three potential weaknesses: (1) it is unclear how t-SNE performs on general dimensionality reduction tasks, (2) the relatively local nature of t-SNE makes it sensitive to the curse of the intrinsic dimensionality of the data, and (3) t-SNE is not guaranteed to converge to a global optimum of its cost function. Below, we discuss the three weaknesses in more detail.

1) Dimensionality reduction for other purposes. It is not obvious how t-SNE will perform on the more general task of dimensionality reduction (i.e., when the dimensionality of the data is not reduced to two or three, but to $d > 3$ dimensions). To simplify evaluation issues, this paper only considers the use of t-SNE for data visualization. The behavior of t-SNE when reducing data to two or three dimensions cannot readily be extrapolated to $d > 3$ dimensions because of the heavy tails of the Student-t distribution. In high-dimensional spaces, the heavy tails comprise a relatively large portion of the probability mass under the Student-t distribution, which might lead to d -dimensional data representations that do not preserve the local structure of the data as well. Hence, for tasks

可以证明，当所有 $(n-1)$ 个非平凡特征维克托尔都被使用时，扩散映射中的欧几里得距离等于高维数据表示中的扩散距离 (Lafon 和 Lee, 2006)。从数学上讲，扩散映射最小化

$$C = \sum_i \sum_j \left(D(x_i, x_j) - \|y_i - y_j\| \right).$$

因此，扩散图容易遇到与经典缩放相同的问题：它们对较大的两两扩散距离的建模比小距离更为重视，因此难以很好地保留数据的局部结构。此外，与随机游走版本的 t-SNE 不同，扩散映射没有自然选择随机游走长度 t 的方法。

在补充材料中，我们展示了 t-SNE 优于 CCA (Demartines 和 H'erault, 1997)、MVU (Weinberger 等, 2004) 和拉普拉斯特征图 (Belkin 和 Niyogi, 2002) 的结果。对于 CCA 及其密切相关的 CDA (Lee 等, 2000)，这些结果部分可以用这些技术定义的局部与全局结构之间的硬边界 λ 来解释，而非 t-SNE 的软边界。此外，在 λ 范围内，CCA 存在与萨蒙映射相同的弱点：它极高地重视建模两个极近数据点之间的距离。

与 t-SNE 类似，MVU (Weinberger 等, 2004) 试图很好地模拟所有小间隔，但 MVU 坚持完美建模（即将其视为约束），单个错误约束可能严重影响 MVU 的性能。当弯曲流形的两个部分在本质流形坐标中相距较远时，就会发生短路。此外，MVU 不尝试建模更远距离结构：它仅根据硬性约束将地图点尽可能拉开，因此与 t-SNE 不同，无法期望在地图中生成合理的大尺度结构。

对于拉普拉斯特征映射，相对于 t-SNE 结果较差的原因是拉普拉斯特征映射与 LLE 具有相同的协方差约束，因此很容易在该约束上作弊。

6.2 弱点

尽管我们已证明 t-SNE 在数据可视化技术中表现优异，但 tSNE 存在三个潜在弱点：(1) 尚不清楚 t-SNE 在一般降维任务中的表现；(2) t-SNE 相对局部的特性使其对数据内在维度的诅咒敏感；(3) t-SNE 不保证收敛到其代价函数的全局最优。下面，我们将更详细地讨论这三个弱点。

1) 用于其他目的的降维。目前尚不清楚 t-SNE 在更一般的任务是降维（即数据的维度不是降为二或三维，而是降为 $d > 3$ 维）。为简化评估问题，本文仅考虑 t-SNE 在数据可视化中的应用。由于学生-t 分布的尾部较重，t-SNE 在将数据还原为二维或三维时的行为无法轻易外推到 $d > 3$ 维。在高维空间中，重尾部在学生-t 分布下占概率质量的较大比例，可能导致 d 维数据表示无法同时保持数据的局部结构。因此，对于任务

in which the dimensionality of the data needs to be reduced to a dimensionality higher than three, Student t-distributions with more than one degree of freedom¹⁰ are likely to be more appropriate.

2) *Curse of intrinsic dimensionality.* t-SNE reduces the dimensionality of data mainly based on local properties of the data, which makes t-SNE sensitive to the curse of the intrinsic dimensionality of the data (Bengio, 2007). In data sets with a high intrinsic dimensionality and an underlying manifold that is highly varying, the local linearity assumption on the manifold that t-SNE implicitly makes (by employing Euclidean distances between near neighbors) may be violated. As a result, t-SNE might be less successful if it is applied on data sets with a very high intrinsic dimensionality (for instance, a recent study by Meytlis and Sirovich (2007) estimates the space of images of faces to be constituted of approximately 100 dimensions). Manifold learners such as Isomap and LLE suffer from exactly the same problems (see, e.g., Bengio, 2007; van der Maaten et al., 2008). A possible way to (partially) address this issue is by performing t-SNE on a data representation obtained from a model that represents the highly varying data manifold efficiently in a number of nonlinear layers such as an autoencoder (Hinton and Salakhutdinov, 2006). Such deep-layer architectures can represent complex nonlinear functions in a much simpler way, and as a result, require fewer datapoints to learn an appropriate solution (as is illustrated for a d -bits parity task by Bengio 2007). Performing t-SNE on a data representation produced by, for example, an autoencoder is likely to improve the quality of the constructed visualizations, because autoencoders can identify highly-varying manifolds better than a local method such as t-SNE. However, the reader should note that it is by definition impossible to fully represent the structure of intrinsically high-dimensional data in two or three dimensions.

3) *Non-convexity of the t-SNE cost function.* A nice property of most state-of-the-art dimensionality reduction techniques (such as classical scaling, Isomap, LLE, and diffusion maps) is the convexity of their cost functions. A major weakness of t-SNE is that the cost function is not convex, as a result of which several optimization parameters need to be chosen. The constructed solutions depend on these choices of optimization parameters and may be different each time t-SNE is run from an initial random configuration of map points. We have demonstrated that the same choice of optimization parameters can be used for a variety of different visualization tasks, and we found that the quality of the optima does not vary much from run to run. Therefore, we think that the weakness of the optimization method is insufficient reason to reject t-SNE in favor of methods that lead to convex optimization problems but produce noticeably worse visualizations. A local optimum of a cost function that accurately captures what we want in a visualization is often preferable to the global optimum of a cost function that fails to capture important aspects of what we want. Moreover, the convexity of cost functions can be misleading, because their optimization is often computationally infeasible for large real-world data sets, prompting the use of approximation techniques (de Silva and Tenenbaum, 2003; Weinberger et al., 2007). Even for LLE and Laplacian Eigenmaps, the optimization is performed using iterative Arnoldi (Arnoldi, 1951) or Jacobi-Davidson (Fokkema et al., 1999) methods, which may fail to find the global optimum due to convergence problems.

7. Conclusions

The paper presents a new technique for the visualization of similarity data that is capable of retaining the local structure of the data while also revealing some important global structure (such as clusters

10. Increasing the degrees of freedom of a Student-t distribution makes the tails of the distribution lighter. With infinite degrees of freedom, the Student-t distribution is equal to the Gaussian distribution.

对于需要将数据维度降低到大于三维的任务，具有多于一自由度的学生 t 分布更为适用。

2) 内在维度的诅咒。t-SNE 主要基于数据的局部性质降低数据的维度，这使得 t-SNE 对数据内在维度的诅咒非常敏感 (Bengio, 2007)。在具有高内在维度且流形变化很大的数据集中，t-SNE 隐含的流形局部线性假设（通过利用邻居间的欧几里得距离）可能会被违反。因此，如果 t-SNE 应用于具有非常高内在维度的数据集，效果可能较差（例如，Meytlis 和 Sirovich (2007) 的最新研究估计面部图像空间约为 100 维）。流形学习器如 Isomap 和 LLE 也面临完全相同的问题（参见例如 Bengio, 2007; van der Maaten 等, 2008）。一种（部分）解决这一问题的可能方法是对从模型中获得的数据表示进行 t-SNE，该模型高效表示多个非线性层（如自编码器）高度变化的数据流形 (Hinton 和 Salakhutdinov, 2006)。这种深层架构可以以更简单的方式表示复杂的非线性函数，因此学习合适解所需的数据点更少（如 Bengio 2007 年在 d 位奇偶校验任务中所示）。例如，对由自编码器生成的数据表示进行 t-SNE 处理，可能会提高构造可视化的质量，因为自编码器比本地方法如 t-SNE 更能识别高度变化流形。然而，读者应注意，从定义上讲，无法完全表示二维或三维中固有高维数据的结构。

3) t-SNE 成本函数的非凸性。大多数最先进的降维技术（如经典缩放、等位图、LLE 和扩散映射）的一个优点是其成本函数的凸性。t-SNE 的一个主要弱点是成本函数不是凸的，因此需要选择多个优化参数。构造的解依赖于这些优化参数的选择，每次从初始随机映射点配置运行 t-SNE 时，解可能都不同。我们已经证明，相同的优化参数选择可以应用于多种不同的可视化任务，并且我们发现最优参数的质量在每次运行中变化不大。因此，我们认为优化方法的弱点不足以成为拒绝 t-SNE 而转而采用导致凸优化问题但可视化效果明显较差的方法的充分理由。准确捕捉可视化中期望的局部最优值通常优于未能捕捉重要方面的成本函数的全局最优值。此外，成本函数的凸性可能具有误导性，因为它们的优化在大型现实数据集中通常计算上不可行，促使采用近似技术 (de Silva 和 Tenenbaum, 2003; Weinberger 等, 2007)。即使是 LLE 和拉普拉斯特征映射，优化也采用迭代 Arnoldi (Arnoldi, 1951) 或 Jacobi-Davidson (Fokkema 等, 1999) 方法，这些方法可能因收敛问题而无法找到全局最优解。

7. 结论

论文提出了一种新的相似度数据可视化技术，能够保留数据的局部结构，同时揭示一些重要的全局结构（如聚类）

10. 增加学生- t 分布的自由度，使分布的尾部变轻。在无限自由度下，学生- t 分布等于高斯分布。

at multiple scales). Both the computational and the memory complexity of t-SNE are $O(n^2)$, but we present a landmark approach that makes it possible to successfully visualize large real-world data sets with limited computational demands. Our experiments on a variety of data sets show that t-SNE outperforms existing state-of-the-art techniques for visualizing a variety of real-world data sets. Matlab implementations of both the normal and the random walk version of t-SNE are available for download at <http://ticc.uvt.nl/~lvdrmaaten/tsne>.

In future work we plan to investigate the optimization of the number of degrees of freedom of the Student-t distribution used in t-SNE. This may be helpful for dimensionality reduction when the low-dimensional representation has many dimensions. We will also investigate the extension of t-SNE to models in which each high-dimensional datapoint is modeled by several low-dimensional map points as in Cook et al. (2007). Also, we aim to develop a parametric version of t-SNE that allows for generalization to held-out test data by using the t-SNE objective function to train a multilayer neural network that provides an explicit mapping to the low-dimensional space.

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Appendix A. Derivation of the t-SNE gradient

t-SNE minimizes the Kullback-Leibler divergence between the joint probabilities p_{ij} in the high-dimensional space and the joint probabilities q_{ij} in the low-dimensional space. The values of p_{ij} are defined to be the symmetrized conditional probabilities, whereas the values of q_{ij} are obtained by means of a Student-t distribution with one degree of freedom

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n},$$

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}},$$

where $p_{j|i}$ and $p_{i|j}$ are either obtained from Equation 1 or from the random walk procedure described in Section 5. The values of p_{ii} and q_{ii} are set to zero. The Kullback-Leibler divergence between the two joint probability distributions P and Q is given by

$$\begin{aligned} C = KL(P||Q) &= \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} \\ &= \sum_i \sum_j p_{ij} \log p_{ij} - p_{ij} \log q_{ij}. \end{aligned} \tag{6}$$

多尺度上)。t-SNE 的计算复杂度和内存复杂度均为 $O(n)$ ，但我们提出了一种具有里程碑意义的方法，使得在有限的计算需求下成功可视化大型现实数据集成为可能。我们在多种数据集上的实验表明，t-SNE 在可视化各种真实世界数据集方面优于现有的最先进技术。t-SNE 的正常版和随机游走版本的 Matlab 实现均可在 <http://ticc.uvt.nl/~lvdrmaaten/tsne> 下载。

未来工作中，我们计划研究 t-SNE 中 Student-t 分布自由度的优化。当低维表示具有多维时，这对降维可能有帮助。我们还将研究 t-SNE 扩展到每个高维数据点由多个低维映射点建模的模型，如 Cook 等人 (2007)。此外，我们还计划开发一种参数化版本的 t-SNE，通过使用 t-SNE 目标函数训练多层神经网络，实现对低维空间的显式映射，从而推广到保留测试数据。

确认

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附录 A。t-SNE 梯度的推导

t-SNE 最小化了高维空间中联合概率与低维空间中联合概率 q_{ij} 之间的 Kullback-Leibler 散度。 p_{ij} 的取值定义为对称化条件概率，而 q_{ij} 的取值则是通过自由度为一的 Student-t 分布得到的

$$p_{ij} = \frac{p_i + p_j}{2n},$$

$$q_{ij} = \frac{\left(1 + \frac{\|x_i - x_j\|^2}{2\sigma^2}\right)^{-1}}{\sum_{k,l} \left(1 + \frac{\|x_k - x_l\|^2}{2\sigma^2}\right)^{-1}},$$

其中 p_i 和 p_{ij} 要么来自方程 1，要么来自第 5 节描述的随机游走过程。 p_i 和 q_i 的值设为零。两个联合概率分布 P 和 Q 之间的 Kullback-Leibler 散度为

$$C = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

$$= \sum_i \sum_j p_{ij} (\log p_{ij} - \log q_{ij}). \quad (6)$$

In order to make the derivation less cluttered, we define two auxiliary variables d_{ij} and Z as follows

$$d_{ij} = \|y_i - y_j\|,$$

$$Z = \sum_{k \neq l} (1 + d_{kl}^2)^{-1}.$$

Note that if y_i changes, the only pairwise distances that change are d_{ij} and d_{ji} for $\forall j$. Hence, the gradient of the cost function C with respect to y_i is given by

$$\begin{aligned} \frac{\delta C}{\delta y_i} &= \sum_j \left(\frac{\delta C}{\delta d_{ij}} + \frac{\delta C}{\delta d_{ji}} \right) (y_i - y_j) \\ &= 2 \sum_j \frac{\delta C}{\delta d_{ij}} (y_i - y_j). \end{aligned} \quad (7)$$

The gradient $\frac{\delta C}{\delta d_{ij}}$ is computed from the definition of the Kullback-Leibler divergence in Equation 6 (note that the first part of this equation is a constant).

$$\begin{aligned} \frac{\delta C}{\delta d_{ij}} &= - \sum_{k \neq l} p_{kl} \frac{\delta(\log q_{kl})}{\delta d_{ij}} \\ &= - \sum_{k \neq l} p_{kl} \frac{\delta(\log q_{kl} Z - \log Z)}{\delta d_{ij}} \\ &= - \sum_{k \neq l} p_{kl} \left(\frac{1}{q_{kl} Z} \frac{\delta((1 + d_{kl}^2)^{-1})}{\delta d_{ij}} - \frac{1}{Z} \frac{\delta Z}{\delta d_{ij}} \right) \end{aligned}$$

The gradient $\frac{\delta((1 + d_{kl}^2)^{-1})}{\delta d_{ij}}$ is only nonzero when $k = i$ and $l = j$. Hence, the gradient $\frac{\delta C}{\delta d_{ij}}$ is given by

$$\frac{\delta C}{\delta d_{ij}} = 2 \frac{p_{ij}}{q_{ij} Z} (1 + d_{ij}^2)^{-2} - 2 \sum_{k \neq l} p_{kl} \frac{(1 + d_{ij}^2)^{-2}}{Z}.$$

Noting that $\sum_{k \neq l} p_{kl} = 1$, we see that the gradient simplifies to

$$\begin{aligned} \frac{\delta C}{\delta d_{ij}} &= 2p_{ij}(1 + d_{ij}^2)^{-1} - 2q_{ij}(1 + d_{ij}^2)^{-1} \\ &= 2(p_{ij} - q_{ij})(1 + d_{ij}^2)^{-1}. \end{aligned}$$

Substituting this term into Equation 7, we obtain the gradient

$$\frac{\delta C}{\delta y_i} = 4 \sum_j (p_{ij} - q_{ij})(1 + \|y_i - y_j\|^2)^{-1} (y_i - y_j).$$

Appendix B. Analytical Solution to Random Walk Probabilities

Below, we describe the analytical solution to the random walk probabilities that are employed in the random walk version of t-SNE (see Section 5). The solution is described in more detail by Grady (2006).

(6) 为了使推导更简洁，我们定义两个辅助变量 d 和 Z ，如下： $d = \|y - y\|$ ，

$$Z = \sum_{\substack{6=1}} (1 + d)。$$

注意，如果 y 变，唯一两两变化的距离是 d and, $j \forall$ 。因此，成本函数 C 相对于 y 的梯度为

$$\begin{aligned} \frac{\delta C}{\delta y} &= \sum \left(\frac{\delta C}{\delta d} + \frac{\delta C}{\delta d} \right) (y - y) \\ &= 2 \sum \frac{C}{\delta d} (y - y)。 \end{aligned} \quad (7)$$

梯度是根据方程 6 中库尔巴克-莱布勒散度的定义计算得出的（注意该方程的第一部分是常数）。

$$\begin{aligned} \frac{\delta C}{\delta d} &= - \sum_{\substack{6=1}} p \frac{\delta (\log q)}{\delta d} \\ &= - \sum_{\substack{6=1}} p \frac{\delta (\log qZ - \log Z)}{\delta d} \\ &= - \sum_{\substack{6=1}} p \left(\frac{1}{qZ} \frac{\delta ((1 + d))}{\delta d} - \frac{1}{Z} \frac{\delta Z}{\delta d} \right) \end{aligned}$$

梯度 只有当 $k = i$ 且 $l = j$ 时， δdis 才非零。因此，梯度由下式给出

$$\frac{\delta C}{\delta d} = 2 \frac{p}{qZ} (1 + d) - 2 \sum_{\substack{6=1}} p \frac{(1 + d)}{Z}。$$

注意 $\sum p = 1$ ，我们看到梯度简化为

$$\begin{aligned} \frac{\delta C}{\delta d} &= 2p (1 + d) - 2q (1 + d) \\ &= 2 (p - q) (1 + d)。 \end{aligned}$$

将该项代入方程 7，得到梯度

$$\frac{\delta C}{\delta y} = 4 \sum (p - q) (1 + \|y - y\|) (y - y)。$$

附录 B。随机游走概率的解析解

下面，我们将描述随机游走版本中所采用的随机游走概率的解析解（见第 5 节）。Grady（2006）对该解有更详细的描述。

It can be shown that computing the probability that a random walk initiated from a non-landmark point (on a graph that is specified by adjacency matrix W) first reaches a specific landmark point is equal to computing the solution to the combinatorial Dirichlet problem in which the boundary conditions are at the locations of the landmark points, the considered landmark point is fixed to unity, and the other landmarks points are set to zero (Kakutani, 1945; Doyle and Snell, 1984). In practice, the solution can thus be obtained by minimizing the combinatorial formulation of the Dirichlet integral

$$D[x] = \frac{1}{2} x^T L x,$$

where L represents the graph Laplacian. Mathematically, the graph Laplacian is given by $L = D - W$, where $D = \text{diag}(\sum_j w_{1j}, \sum_j w_{2j}, \dots, \sum_j w_{nj})$. Without loss of generality, we may reorder the landmark points such that the landmark points come first. As a result, the combinatorial Dirichlet integral decomposes into

$$\begin{aligned} D[x_N] &= \frac{1}{2} \begin{bmatrix} x_L^T & x_N^T \end{bmatrix} \begin{bmatrix} L_L & B \\ B^T & L_N \end{bmatrix} \begin{bmatrix} x_L \\ x_N \end{bmatrix} \\ &= \frac{1}{2} (x_L^T L_L x_L + 2x_N^T B^T x_L + x_N^T L_N x_N), \end{aligned}$$

where we use the subscript $\cdot_L$ to indicate the landmark points, and the subscript $\cdot_N$ to indicate the non-landmark points. Differentiating $D[x_N]$ with respect to x_N and finding its critical points amounts to solving the linear systems

$$L_N x_N = -B^T. \quad (8)$$

Please note that in this linear system, B^T is a matrix containing the columns from the graph Laplacian L that correspond to the landmark points (excluding the rows that correspond to landmark points). After normalization of the solutions to the systems X_N , the column vectors of X_N contain the probability that a random walk initiated from a non-landmark point terminates in a landmark point. One should note that the linear system in Equation 8 is only nonsingular if the graph is completely connected, or if each connected component in the graph contains at least one landmark point (Biggs, 1974).

Because we are interested in the probability of a random walk initiated from a *landmark point* terminating at another landmark point, we duplicate all landmark points in the neighborhood graph, and initiate the random walks from the duplicate landmarks. Because of memory constraints, it is not possible to store the entire matrix X_N into memory (note that we are only interested in a small number of rows from this matrix, viz., in the rows corresponding to the duplicate landmark points). Hence, we solve the linear systems defined by the columns of $-B^T$ one-by-one, and store only the parts of the solutions that correspond to the duplicate landmark points. For computational reasons, we first perform a Cholesky factorization of L_N , such that $L_N = CC^T$, where C is an upper-triangular matrix. Subsequently, the solution to the linear system in Equation 8 is obtained by solving the linear systems $Cy = -B^T$ and $Cx_N = y$ using a fast backsubstitution method.

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可以证明，计算从非地标点（在由邻接矩阵 W 指定图上）开始的随机游走首先到达特定地标点的概率，等于计算组合狄利克雷问题的解，其中边界条件位于地标点的位置，考虑的地标点固定为一，其他地标点设为零（Kakutani，1945;Doyle 和 Snell，1984）。实际上，可以通过最小化狄利克雷积分的组合表述来获得该解

$$D[x] = \frac{1}{2}xLx,$$

其中 L 表示拉普拉斯算子图。数学上，拉普拉斯算子由 $L = D - W$ 给出，其中 $D = \text{diag}$ 。不失一般性，我们可以重新排序地标点，使得地标点先出现。因此，组合狄利克雷积分分解为

$$\begin{aligned} D[x] &= \frac{1}{2} \begin{bmatrix} x & x \\ & \end{bmatrix} \begin{bmatrix} L & B \\ B^T & L \end{bmatrix} \begin{bmatrix} x \\ x \end{bmatrix} \\ &= \frac{1}{2} (xLx + 2xBx + xLx), \end{aligned}$$

其中我们使用下标·表示地标点，下标·表示非地标点。对 $x[x]$ 微分并找到临界点，就是求解线性系统

$$Lx = -B. \tag{8}$$

请注意，在该线性系统中，B 是一个矩阵，包含拉普拉斯图 L 中对应地标点的列（不包括对应地标点的行）。在对系统 X 的解进行归一化后，X 的列向量包含从非里程碑点开始的随机游走终止于某个里程碑点的概率。应注意，方程 8 中的线性系统只有在图完全连通，或图中每个连通分量至少包含一个标志点时才非奇异（Biggs，1974）。
由于我们关注从一个地标点开始的随机游走终止于另一个地标点的概率，我们对邻域图中的所有地标点进行重复，并从重复的地标开始随机游走。由于内存限制，无法将整个矩阵 X 存储到内存中（注意我们只关注该矩阵中少量行，即对应重复标志点的行）。因此，我们求解由 Bone-by-one 列定义的线性系统，并仅存储对应重复地标点的解部分。出于计算原因，我们首先对 L 进行乔莱斯基分解，使得 $L = CC^T$ ，其中 C 是一个上三角矩阵。随后，通过快速回代法求解线性系统 $Cy = -B$ ，即可得到方程 8 中的线性系统解。

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